

Project: Reactjs E-commerce Application

1.Docker:

1.Docker and Docker-compose Installed:

```
root@ip-172-31-24-31:~# docker --version && docker-compose --version
Docker version 27.5.1, build 27.5.1-0ubuntu3~24.04.2
docker-compose version 1.29.2, build unknown
root@ip-172-31-24-31:~#
```

2.Source Code File:

```
root@ip-172-31-24-31:~/Project1# ls
build
root@ip-172-31-24-31:~/Project1#
```

3.Creating Dockerfile:

```
root@ip-172-31-24-31:~/Project1# nano Dockerfile
```

4.DockerFile Code:

```
GNU nano 7.2 Dockerfile
FROM nginx:alpine

RUN rm -rf /usr/share/nginx/html/*

COPY build/ /usr/share/nginx/html/

EXPOSE 80

CMD ["nginx", "-g", "daemon off;"]
```

5.Creating Docker Image:

```
root@ip-172-31-24-31:~/Project1# docker build -t static-app .
DEPRECATED: The legacy builder is deprecated and will be removed in a future release.
            Install the buildx component to build images with BuildKit:
            https://docs.docker.com/go/buildx/

Sending build context to Docker daemon  3.419MB
Step 1/6 : FROM nginx:alpine
alpine: Pulling from library/nginx
9824c27679d3: Pull complete
6bc572a340ec: Pull complete
403e3f251637: Pull complete
9adfb9e99cb7: Pull complete
7a8a46741e18: Pull complete
c9ebe2ff2d2c: Pull complete
a992fbc61ecc: Pull complete
cb1ff4086f82: Pull complete
Digest: sha256:42a516af16b852e33b7682d5ef8acbd5d13fe08fecadc7ed98605ba5e3b26ab8
Status: Downloaded newer image for nginx:alpine
--> 4a86014ec699
Step 2/6 : WORKDIR /usr/share/nginx/html
--> Running in df7b442476fa
--> Removed intermediate container df7b442476fa
--> afbdd68e0ffd
Step 3/6 : RUN rm -rf ./*
```

6.Docker Image Created:

```
root@ip-172-31-24-31:~/Project1# docker images
REPOSITORY    TAG       IMAGE ID       CREATED        SIZE
static-app    latest    96a05b569493   About a minute ago  55.9MB
nginx         alpine    4a86014ec699   9 days ago     52.5MB
root@ip-172-31-24-31:~/Project1#
```

7.Docker-compose Code:

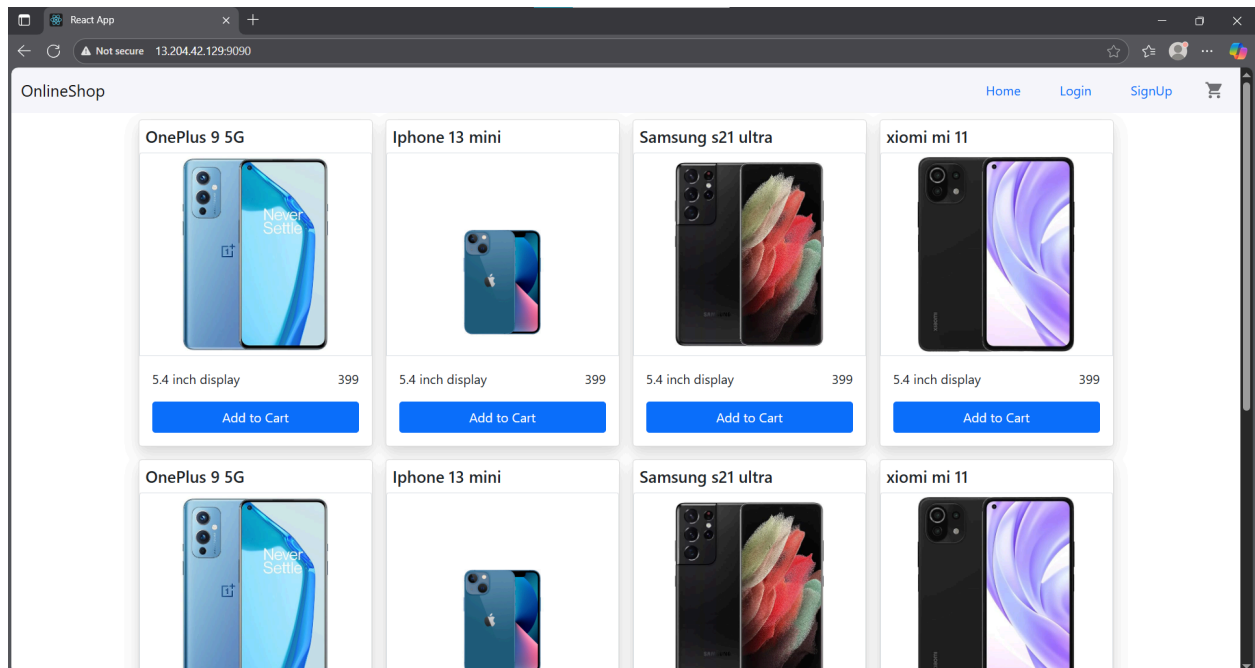
```
GNU nano 7.2                                docker-compose.yaml
version: '3.8'

services:
  Dev:
    image: static-app:latest
    ports:
      - "9090:80"
```

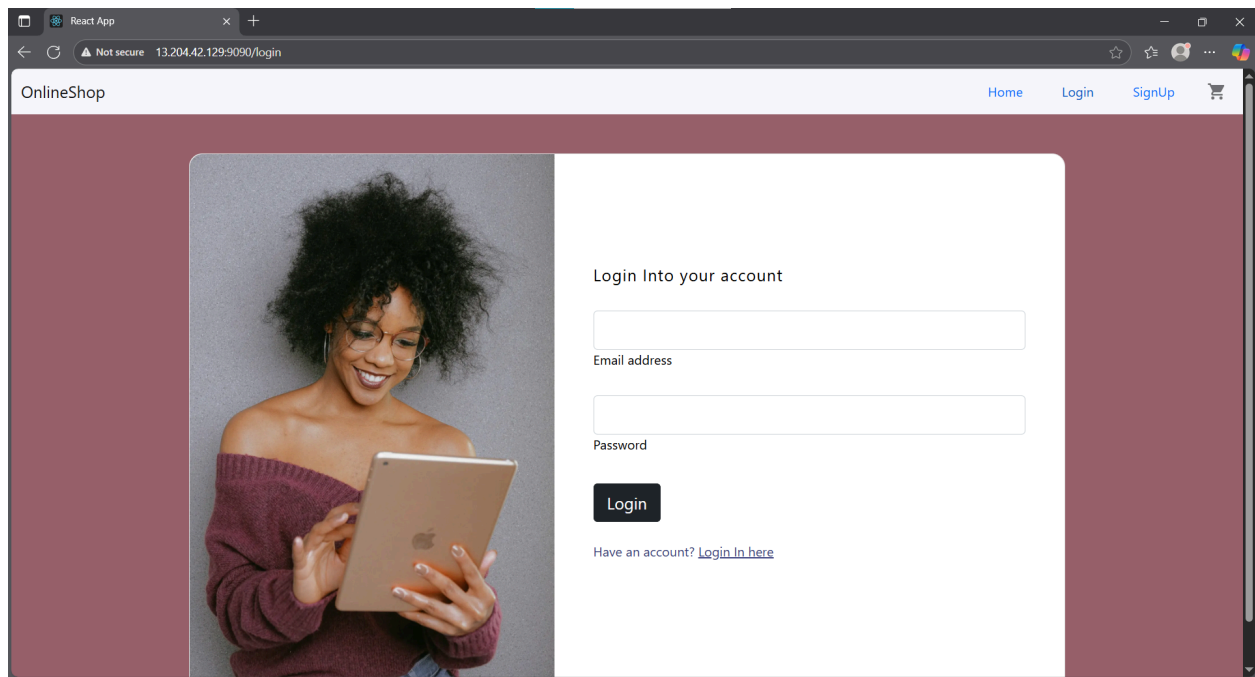
8.Running The Docker-compose:

```
root@ip-172-31-24-31:~/Project1# docker-compose up -d
Creating network "project1_default" with the default driver
Creating project1_Dev_1 ... done
root@ip-172-31-24-31:~/Project1#
```

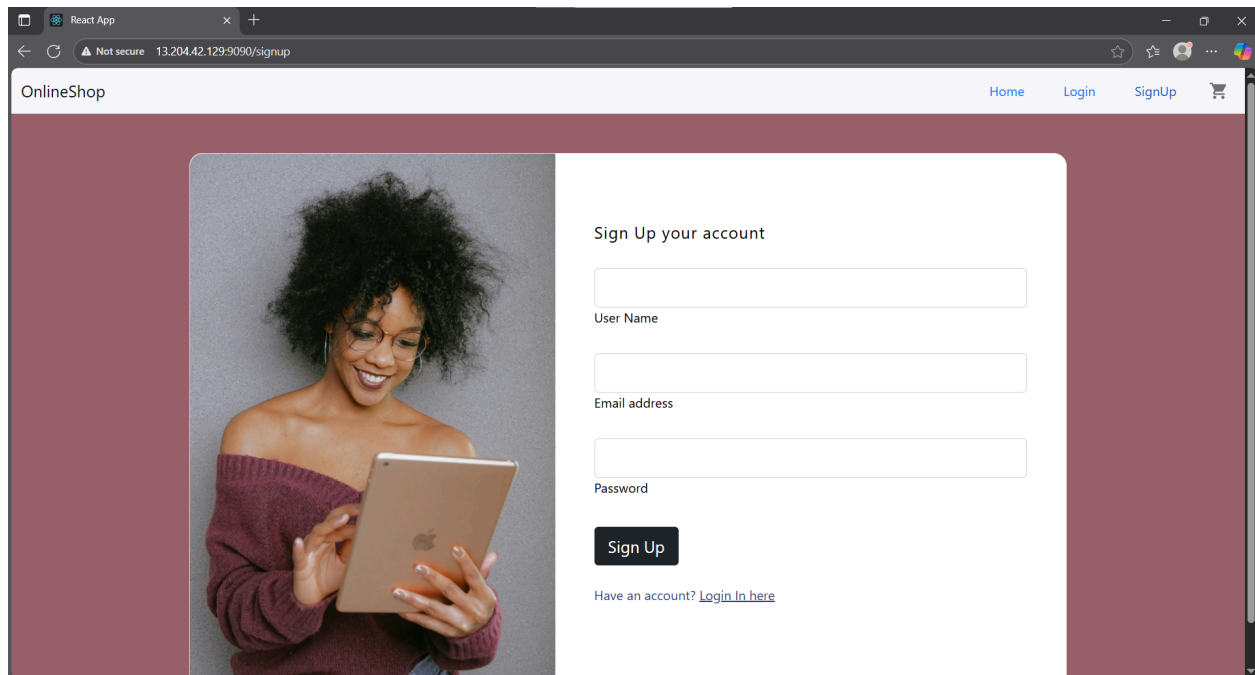
9. Docker Application Output 1:



10. Docker Application Output 2:



11.Docker Application Output 3:



2.Bash_Scripting:

1.Creating Script file to Build Docker Image:

```
GNU nano 7.2                               build.sh *
#!/bin/bash

IMAGE_NAME="static-web"

docker build -t $IMAGE_NAME .
```

2.Creating Script file to Run Docker Container:

```
GNU nano 7.2                                deploy.sh *
#!/bin/bash

IMAGE_NAME="static-web"

CONTAINER_NAME="static_application"

docker rm -f $CONTAINER_NAME 2>/dev/null || true

docker run -idt -p "9091:80" --name $CONTAINER_NAME $IMAGE_NAME
```

3.Using Docker Ignore to ignore some files in Docker Build:

```
GNU nano 7.2                                .dockerignore *
*.sh
*.yaml
```

4.Giving Executable permission to Build and Deploy file:

```
root@ip-172-31-24-31:~/Project1# chmod +x build.sh deploy.sh
root@ip-172-31-24-31:~/Project1#
```

5. Running Build File:

```
root@ip-172-31-24-31:~/Project1# ./build.sh
DEPRECATED: The legacy builder is deprecated and will be removed in a future release.
            Install the buildx component to build images with BuildKit:
            https://docs.docker.com/go/buildx/

Sending build context to Docker daemon   3.42MB
Step 1/5 : FROM nginx:alpine
--> 4a86014ec699
Step 2/5 : RUN rm -rf /usr/share/nginx/html/*
--> Using cache
--> dcf90923529c
Step 3/5 : COPY build/ /usr/share/nginx/html/
--> Using cache
--> 8ae700e55a18
Step 4/5 : EXPOSE 80
--> Using cache
--> 8d976824691c
Step 5/5 : CMD ["nginx", "-g", "daemon off;"]
--> Using cache
--> 836000789d26
Successfully built 836000789d26
Successfully tagged static-web:latest
root@ip-172-31-24-31:~/Project1#
```

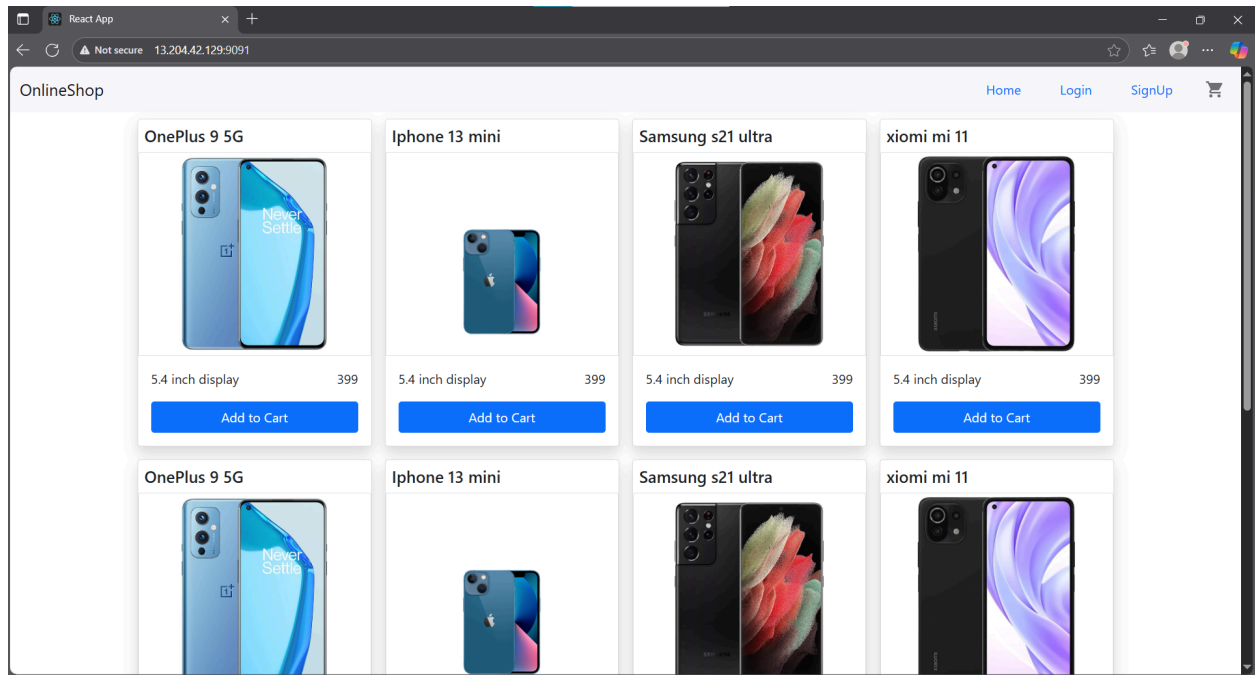
6. Docker Image created:

```
root@ip-172-31-24-31:~/Project1# docker images
REPOSITORY    TAG       IMAGE ID       CREATED        SIZE
static-web    latest    fded10026b8c   17 seconds ago 55.1MB
nginx         alpine    4a86014ec699   9 days ago    52.5MB
root@ip-172-31-24-31:~/Project1#
```

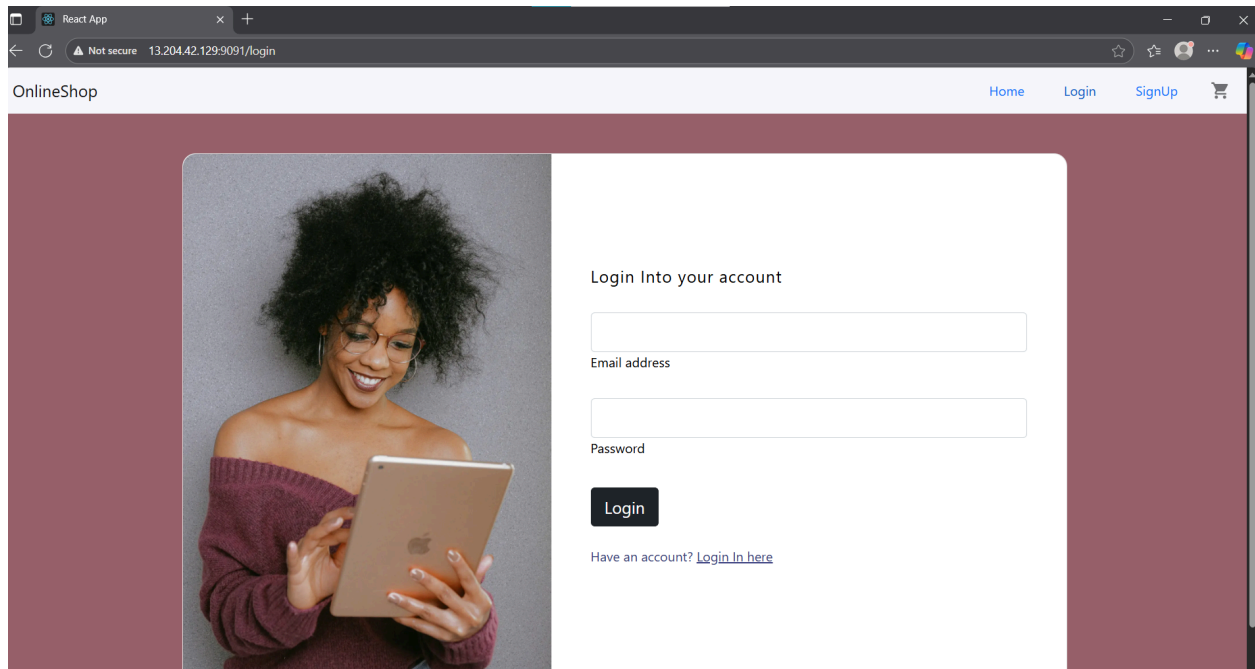
7. Docker Container Created:

```
root@ip-172-31-24-31:~/Project1# ./deploy.sh
76e6a35d87d1fead112387509a039673b2892befbbce524b03231cd2fff63f72
root@ip-172-31-24-31:~/Project1# docker ps
CONTAINER ID   IMAGE      COMMAND                  CREATED        STATUS        PORTS                               NAMES
76e6a35d87d1   static-web "/docker-entrypoint..." 7 seconds ago  Up 6 seconds  0.0.0.0:9091->80/tcp, [::]:9091->80/tcp  static_application
root@ip-172-31-24-31:~/Project1#
```

8.Output 1:



9.Output 2:



10.Output 3:

React App

Not secure 13.204.42.129:9091/signup

OnlineShop [Home](#) [Login](#) [SignUp](#)



Sign Up your account

User Name

Email address

Password

[Sign Up](#)

Have an account? [Login In here](#)

3.Version Control:

1.Creating GitHub Repository:

Create a new repository [Preview](#) [Switch back to classic experience](#)

Repositories contain a project's files and version history. Have a project elsewhere? [Import a repository](#).
Required fields are marked with an asterisk (*).

1 General

Owner * Anand-kumar-git / Repository name * Project1

Project1 is available.

Great repository names are short and memorable. How about [super-duper-octo-rotary-phone](#)?

Description

0 / 350 characters

2 Configuration

Choose visibility * Public

Choose who can see and commit to this repository

Add README Off

READMEs can be used as longer descriptions. [About READMEs](#)

Add .gitignore No .gitignore

[.gitignore](#) tells git which files not to track. [About ignoring files](#)

2.Cloning the git Repository:

```
ADMIN@DESKTOP-0E234HV MINGW64 ~/Desktop/Devops/Project1 (master)
• $ git init
Initialized empty Git repository in C:/Users/ADMIN/Desktop/Devops/Project1/.git/

ADMIN@DESKTOP-0E234HV MINGW64 ~/Desktop/Devops/Project1 (master)
• $ git remote add origin https://github.com/Anand-kumar-git/Project1.git

ADMIN@DESKTOP-0E234HV MINGW64 ~/Desktop/Devops/Project1 (master)
• $ git branch -M main

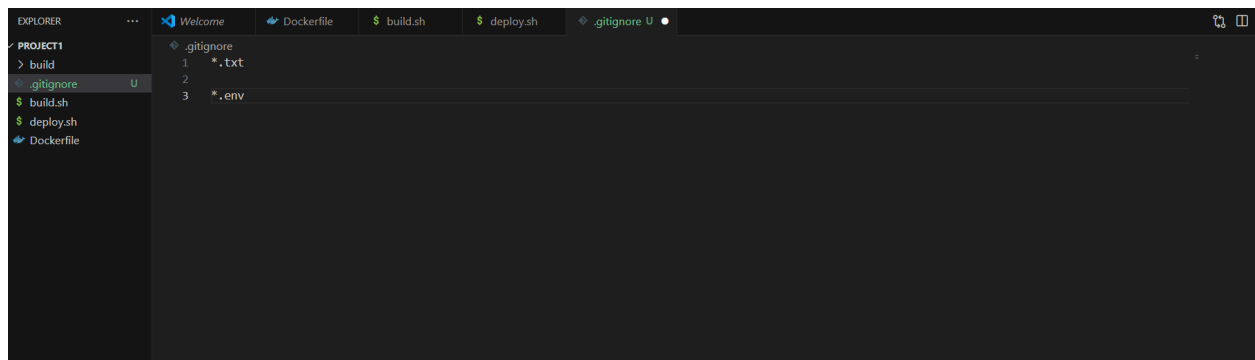
ADMIN@DESKTOP-0E234HV MINGW64 ~/Desktop/Devops/Project1 (main)
○ $
```

3.Creating Dev Branch:

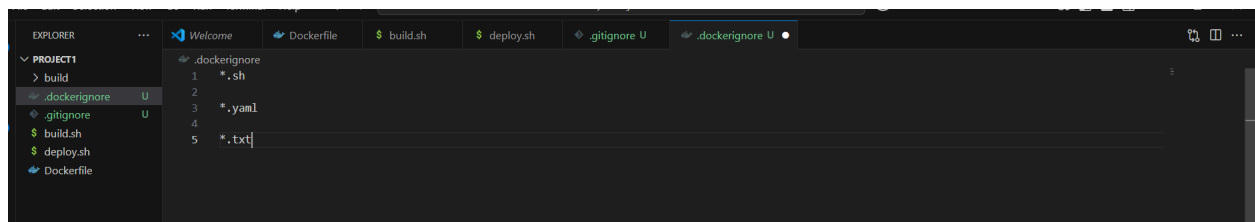
```
ADMIN@DESKTOP-0E234HV MINGW64 ~/Desktop/Devops/Project1 (main)
$ git checkout -b dev
Switched to a new branch 'dev'

ADMIN@DESKTOP-0E234HV MINGW64 ~/Desktop/Devops/Project1 (dev)
$
```

4.Using Git ignore file:



5.Using .dockerignore file:



4.Docker_Hub:

1.Creating Dev Repository in Dockerhub:

anand20003
Docker Personal

Repositories

Collaborations

Settings

Default privacy

Notifications

Billing

Usage

Pulls

Storage

Repositories / Create

Create repository

Repository Name *
dev

Short description

A short description to identify your repository. If the repository is public, this description is used to index your content on Docker Hub and in search engines, and is visible to users in search results.

Visibility

Using 0 of 1 private repositories. [Get more](#)

☒ Public

Appears in Docker Hub search results

☐ Private

Only visible to you

Cancel

Create

Pushing images

You can push a new image to this repository using the CLI:

```
docker tag local-image:tagname new-repo:tagname  
docker push new-repo:tagname
```

Make sure to replace `tagname` with your desired image repository tag.

Using 0 of 1 private repositories. [Get more](#)

2.Creating Prod Private Repository:

anand20003
Docker Personal

Repositories

Collaborations

Settings

Default privacy

Notifications

Billing

Usage

Pulls

Storage

Repositories / Create

Create repository

Repository Name *
prod

Short description

A short description to identify your repository. If the repository is public, this description is used to index your content on Docker Hub and in search engines, and is visible to users in search results.

Visibility

Using 0 of 1 private repositories. [Get more](#)

☐ Public

Appears in Docker Hub search results

☒ Private

Only visible to you

Cancel

Create

Pushing images

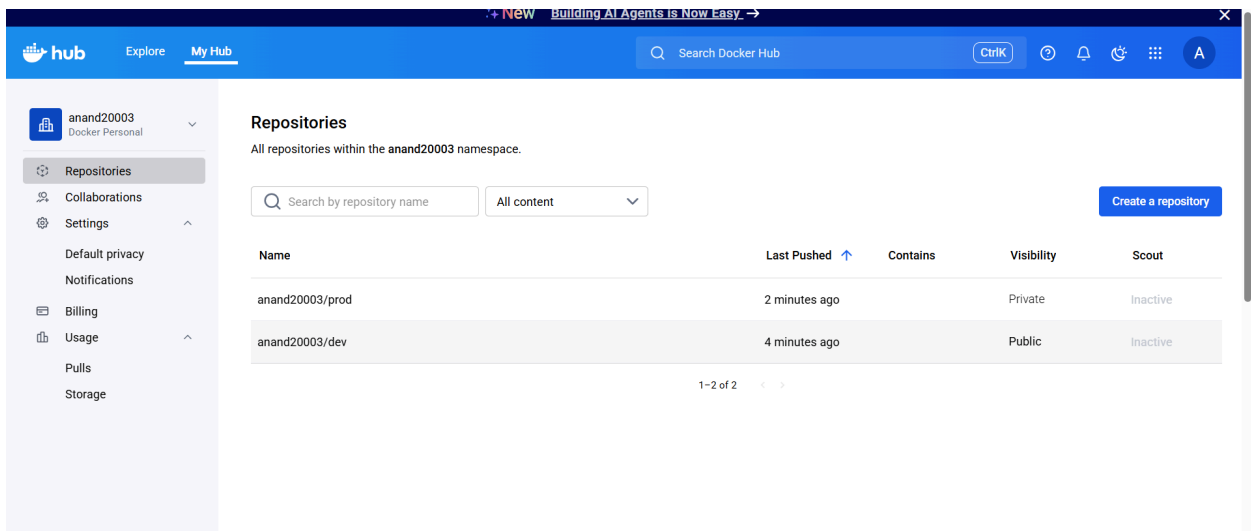
You can push a new image to this repository using the CLI:

```
docker tag local-image:tagname new-repo:tagname  
docker push new-repo:tagname
```

Make sure to replace `tagname` with your desired image repository tag.

Using 0 of 1 private repositories. [Get more](#)

3.Dev and Prod Repository Created:



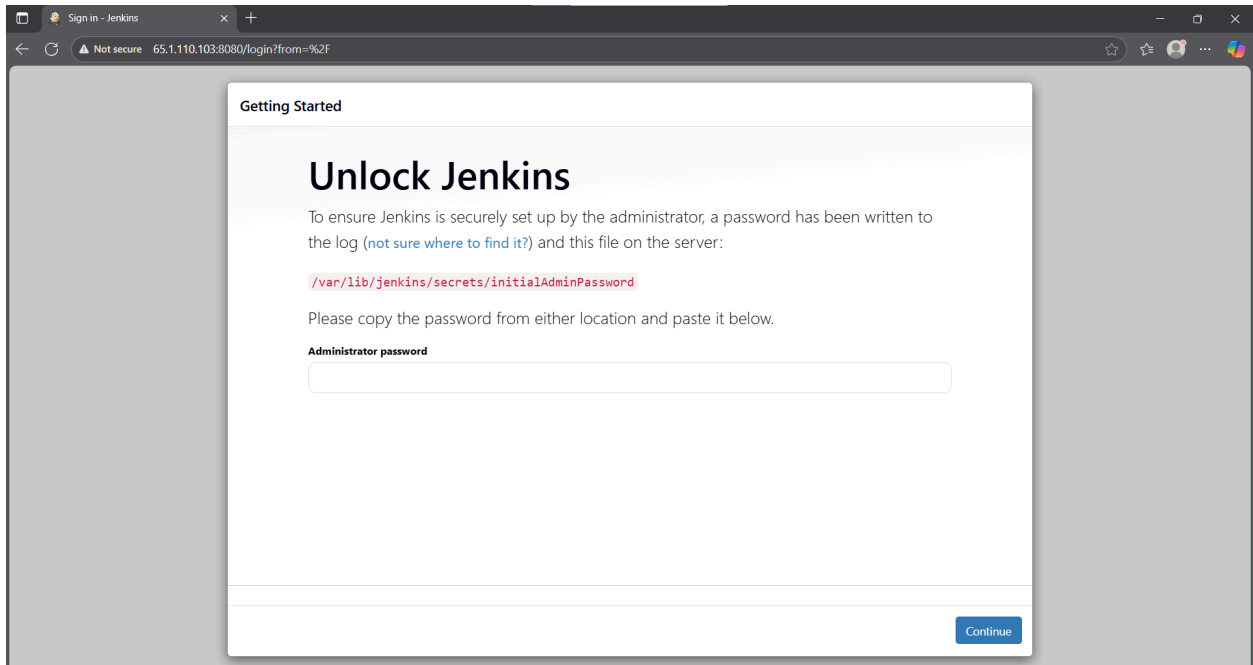
5.Jenkins:

1.Jenkins Installed:

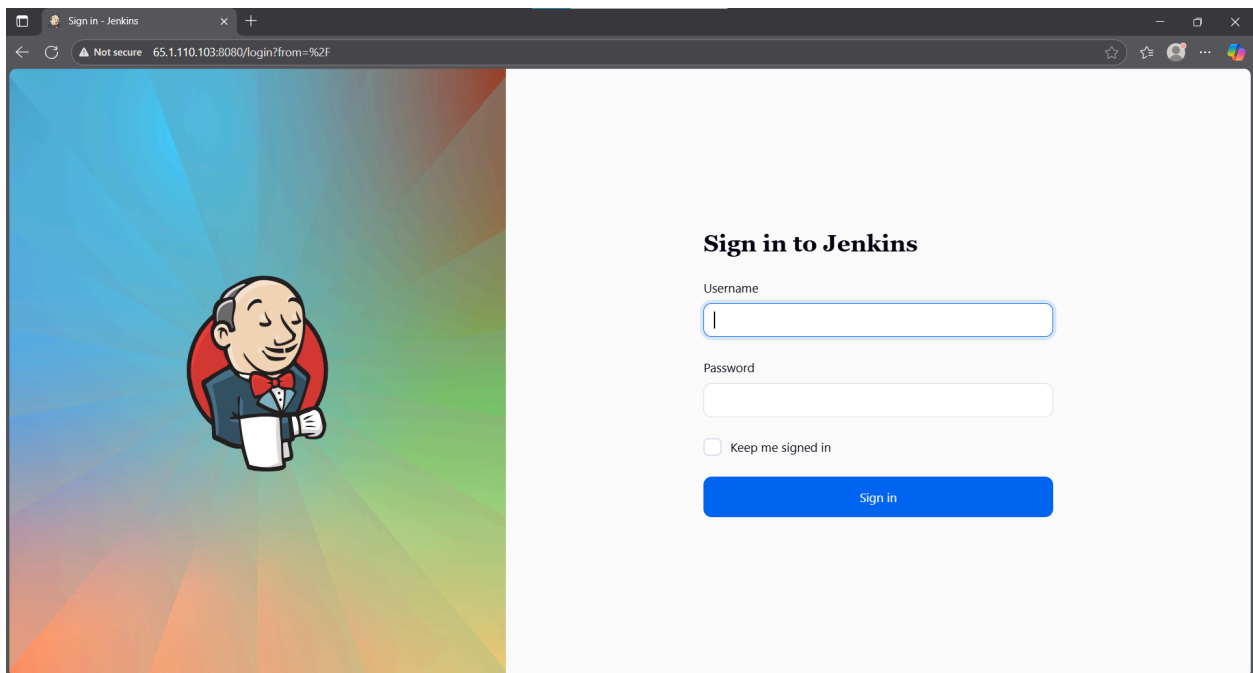
```
root@ip-172-31-28-7:~# systemctl status jenkins
● jenkins.service - Jenkins Continuous Integration Server
   Loaded: loaded (/usr/lib/systemd/system/jenkins.service; enabled; preset: enabled)
   Active: active (running) since Sat 2025-08-23 09:19:53 UTC; 1min 37s ago
     Main PID: 4754 (java)
        Tasks: 45 (limit: 4515)
      Memory: 618.5M (peak: 632.3M)
         CPU: 23.118s
    CGroup: /system.slice/jenkins.service
            └─4754 /usr/bin/java -Djava.awt.headless=true -jar /usr/share/java/jenkins.war --webroot=/var/cache/jenkins/war --httpPort=8080

Aug 23 09:19:46 ip-172-31-28-7 jenkins[4754]: cd3d0ca35989445680e5bb27639bd7a8
Aug 23 09:19:46 ip-172-31-28-7 jenkins[4754]: This may also be found at: /var/lib/jenkins/secrets/initialAdminPassword
Aug 23 09:19:46 ip-172-31-28-7 jenkins[4754]: *****
Aug 23 09:19:46 ip-172-31-28-7 jenkins[4754]: *****
Aug 23 09:19:46 ip-172-31-28-7 jenkins[4754]: *****
Aug 23 09:19:53 ip-172-31-28-7 jenkins[4754]: 2025-08-23 09:19:53.063+0000 [id=33] INFO jenkins.InitReactorRunner$1#onAttained: Completed initialization
Aug 23 09:19:53 ip-172-31-28-7 jenkins[4754]: 2025-08-23 09:19:53.082+0000 [id=23] INFO hudson.lifecycle.Lifecycle#onReady: Jenkins is fully up and running
Aug 23 09:19:53 ip-172-31-28-7 systemd[1]: Started jenkins.service - Jenkins Continuous Integration Server.
Aug 23 09:19:54 ip-172-31-28-7 jenkins[4754]: 2025-08-23 09:19:54.682+0000 [id=49] INFO h.m.DownloadService$Downloadable#load: Obtained the updated data
Aug 23 09:19:54 ip-172-31-28-7 jenkins[4754]: 2025-08-23 09:19:54.683+0000 [id=49] INFO hudson.util.Retrier#start: Performed the action check updates see
lines 1-20/20 (END)
```

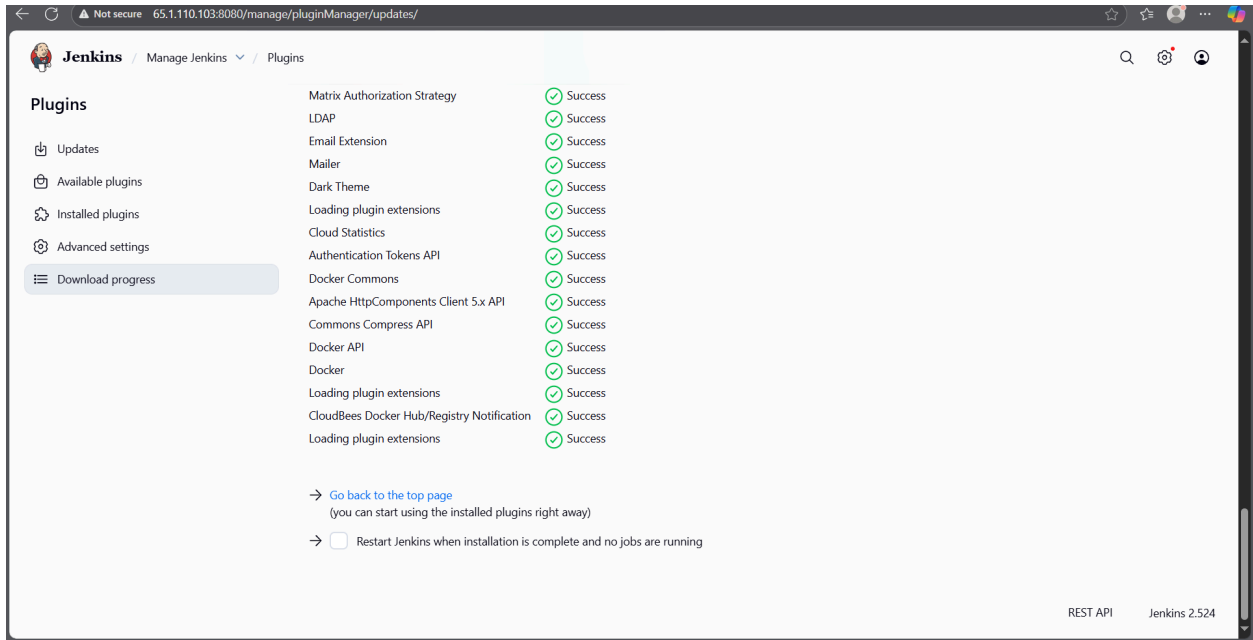
2. Accessing Jenkins With Port 8080:



3. Jenkins Login Page:



4.Installing Required plugins:



The screenshot shows the Jenkins 'Manage Jenkins' > 'Plugins' page. A table lists 20 plugins, all of which have been successfully installed, indicated by green checkmarks and the word 'Success'.

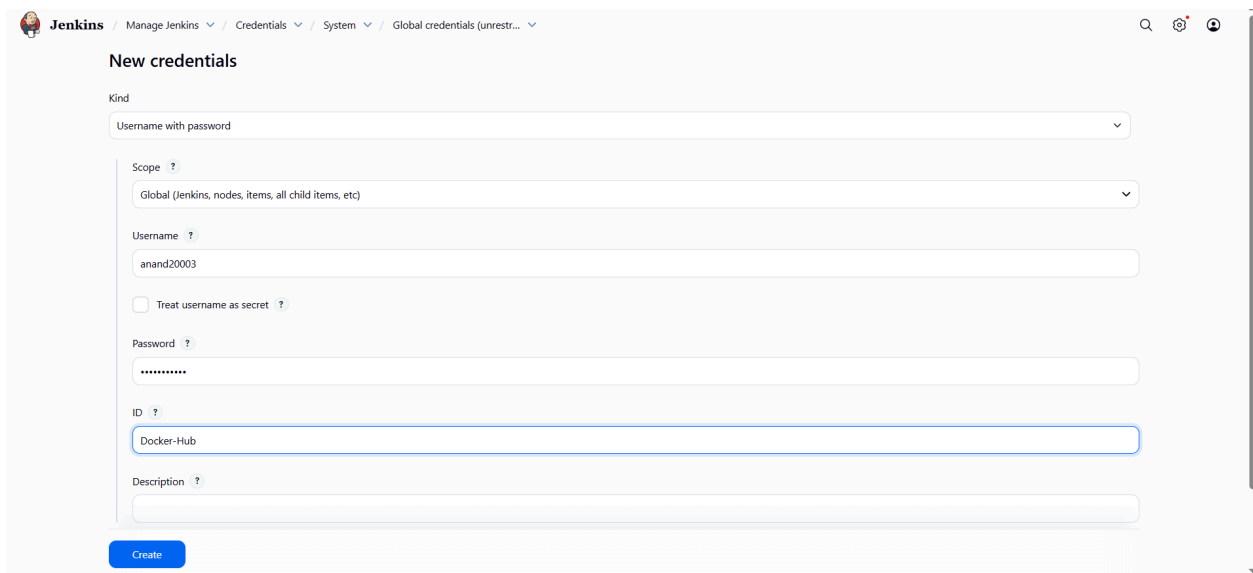
Plugin Name	Status
Matrix Authorization Strategy	Success
LDAP	Success
Email Extension	Success
Mailer	Success
Dark Theme	Success
Loading plugin extensions	Success
Cloud Statistics	Success
Authentication Tokens API	Success
Docker Commons	Success
Apache HttpComponents Client 5.x API	Success
Commons Compress API	Success
Docker API	Success
Docker	Success
Loading plugin extensions	Success
CloudBees Docker Hub/Registry Notification	Success
Loading plugin extensions	Success

Below the table, there are two links:

- [Go back to the top page](#) (you can start using the installed plugins right away)
- ☐ Restart Jenkins when installation is complete and no jobs are running

The bottom right corner of the page shows 'REST API' and 'Jenkins 2.524'.

5.Adding DockerHub Credential in Jenkins:



The screenshot shows the 'New credentials' form in Jenkins. The form is for creating a 'Username with password' credential with a 'Global' scope. The username is 'anand20003' and the ID is 'Docker-Hub'.

Kind: Username with password

Scope: Global (Jenkins, nodes, items, all child items, etc)

Username: anand20003

☐ Treat username as secret

Password: [Redacted]

ID: Docker-Hub

Description: [Empty]

Create

6. Creating Jenkins MultiBranch Pipeline:

← → ↻ Not secure 3.111.38.174:8080/view/all/newJob

25.7: Worked Exam...

Jenkins / All / New Item

New Item

Enter an item name

Project1

Select an item type

- Freestyle project**
Classic, general-purpose job type that checks out from up to one SCM, executes build steps serially, followed by post-build steps like archiving artifacts and sending email notifications.
- Pipeline**
Orchestrates long-running activities that can span multiple build agents. Suitable for building pipelines (formerly known as workflows) and/or organizing complex activities that do not easily fit in free-style job type.
- Multi-configuration project**
Suitable for projects that need a large number of different configurations, such as testing on multiple environments, platform-specific builds, etc.
- Folder**
Creates a container that stores nested items in it. Useful for grouping things together. Unlike view, which is just a filter, a folder creates a separate namespace, so you can have multiple things of the same name as long as they are in different folders.
- Multibranch Pipeline**
Creates a set of Pipeline projects according to detected branches in one SCM repository.

OK

7. Multibranch Pipeline Configuration:

25.7: Worked Exam...

Jenkins / Project1 / Configuration

Configuration

- General
- Branch Sources
- Build Configuration
- Scan Multibranch Pipeline Triggers
- Orphaned Item Strategy
- Appearance
- Health metrics
- Properties

Branch Sources

Git

Project Repository ?

https://github.com/Anand-kumar-git/Project1.git

Credentials ?

- none -

+ Add

Behaviors

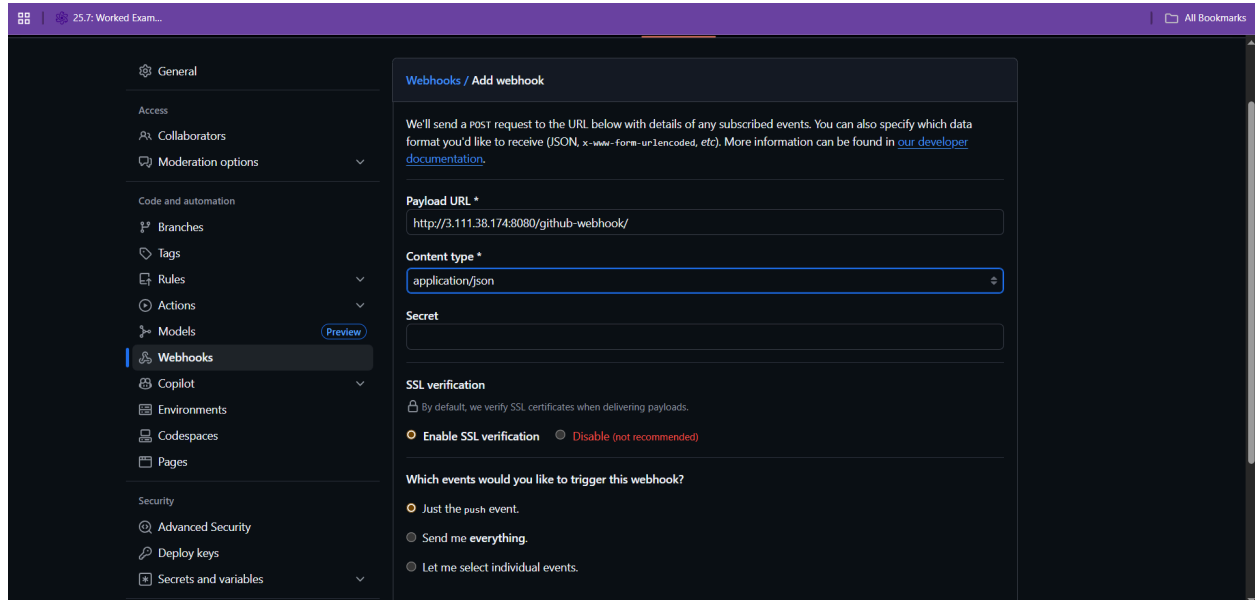
Discover branches

Add

Property strategy

Save Apply

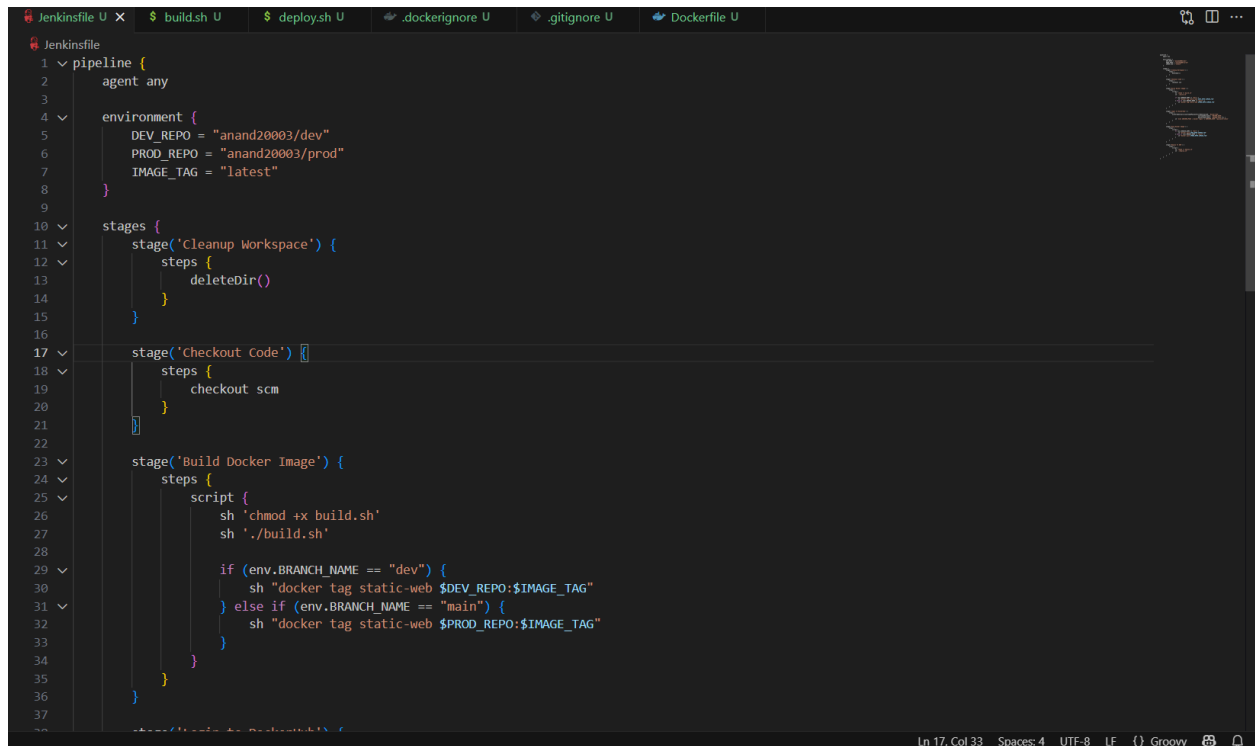
8.Adding Github Webhook:



The screenshot shows the Jenkins 'Webhooks / Add webhook' configuration page. The left sidebar contains a navigation menu with categories: General, Access, Code and automation, and Security. The 'Webhooks' option is selected. The main content area is titled 'Webhooks / Add webhook' and contains the following fields and options:

- Payload URL ***: A text input field containing 'http://3.111.38.174:8080/github-webhook/'.
- Content type ***: A dropdown menu set to 'application/json'.
- Secret**: An empty text input field.
- SSL verification**: A section with a note 'By default, we verify SSL certificates when delivering payloads.' and two radio buttons: 'Enable SSL verification' (selected) and 'Disable (not recommended)'.
- Which events would you like to trigger this webhook?**: A section with three radio buttons: 'Just the push event.' (selected), 'Send me everything.', and 'Let me select individual events.'

9.Jenkinsfile First part:



The screenshot shows a Jenkinsfile script in a code editor. The script is written in Groovy and defines a pipeline with the following structure:

```
1 pipeline {
2   agent any
3
4   environment {
5     DEV_REPO = "anand20003/dev"
6     PROD_REPO = "anand20003/prod"
7     IMAGE_TAG = "latest"
8   }
9
10  stages {
11    stage('Cleanup Workspace') {
12      steps {
13        deleteDir()
14      }
15    }
16
17    stage('Checkout Code') {
18      steps {
19        checkout scm
20      }
21    }
22
23    stage('Build Docker Image') {
24      steps {
25        script {
26          sh 'chmod +x build.sh'
27          sh './build.sh'
28
29          if (env.BRANCH_NAME == "dev") {
30            sh "docker tag static-web $DEV_REPO:$IMAGE_TAG"
31          } else if (env.BRANCH_NAME == "main") {
32            sh "docker tag static-web $PROD_REPO:$IMAGE_TAG"
33          }
34        }
35      }
36    }
37  }
38}
```

The code editor shows line numbers from 1 to 38. The status bar at the bottom indicates 'Ln 17, Col 33', 'Spaces: 4', 'UTF-8', 'LF', and 'Groovy'.

10. Jenkinsfile Second part:

```
38 stage('login to DockerHub') {
39     steps {
40         withCredentials([usernamePassword(credentialsId: 'Docker-Hub',
41                                         usernameVariable: 'DOCKER_USER',
42                                         passwordVariable: 'DOCKER_PASS')]) {
43             sh 'echo $DOCKER_PASS | docker login -u $DOCKER_USER --password-stdin'
44         }
45     }
46 }
47
48 stage('Push Docker Image') {
49     steps {
50         script {
51             if (env.BRANCH_NAME == "dev") {
52                 sh "docker push $DEV_REPO:$IMAGE_TAG"
53             } else if (env.BRANCH_NAME == "main") {
54                 sh "docker push $PROD_REPO:$IMAGE_TAG"
55             }
56         }
57     }
58 }
59
60 stage('Deploy To AWS') {
61     steps {
62         script {
63             sh 'chmod +x deploy.sh'
64             sh './deploy.sh'
65         }
66     }
67 }
68 }
69
70 }
```

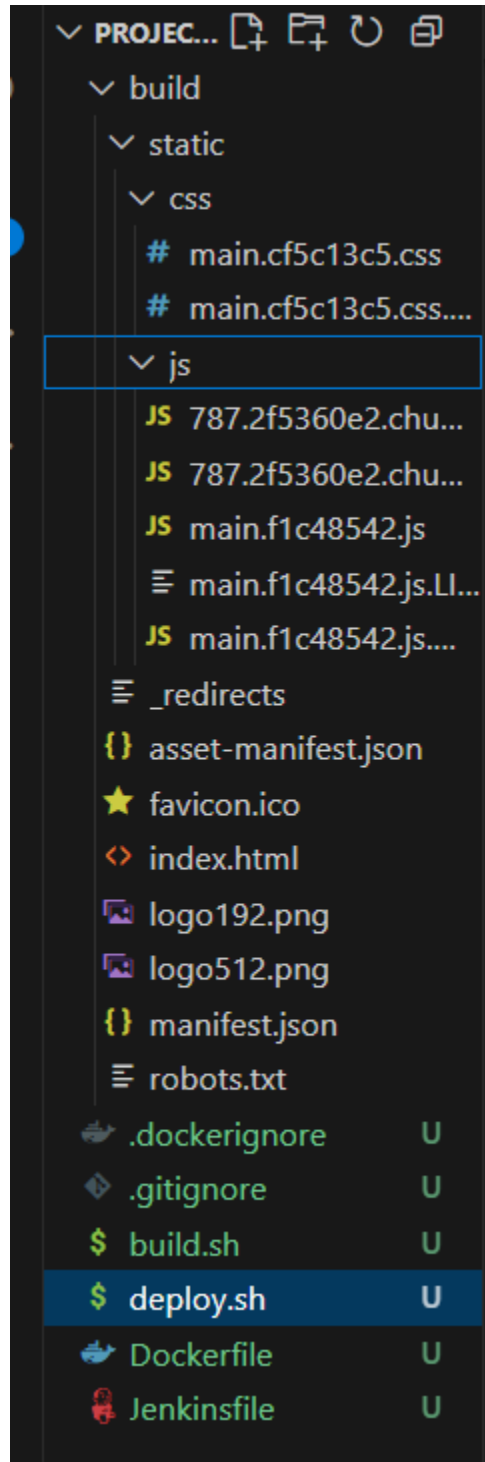
Ln 45, Col 14 Spaces: 4 UTF-8 LF {} Groovy

11. Dockerfile:

```
Dockerfile
1 FROM nginx:alpine
2
3 RUN rm -rf /usr/share/nginx/html/*
4
5 # Copy only prebuilt React app
6 COPY build/. /usr/share/nginx/html/
7
8 EXPOSE 80
9 CMD ["nginx", "-g", "daemon off;"]
10 |
```

Ln 10, Col 1 Spaces: 4 UTF-8 LF {} Docker

12.File Structure:



13.Adding Jenkins to Docker Group:

```
root@ip-172-31-28-7:~# usermod -aG docker jenkins
root@ip-172-31-28-7:~#
```

14.Pushed to dev Branch:

```
ADMIN@DESKTOP-0E234HV MINGW64 ~/Project1 (dev)
• $ git commit -m "First Commit"
[dev b1a4281] First Commit
 1 file changed, 1 insertion(+), 1 deletion(-)

ADMIN@DESKTOP-0E234HV MINGW64 ~/Project1 (dev)
• $ git push
Enumerating objects: 5, done.
Counting objects: 100% (5/5), done.
Delta compression using up to 4 threads
Compressing objects: 100% (3/3), done.
Writing objects: 100% (3/3), 288 bytes | 288.00 KiB/s, done.
Total 3 (delta 2), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (2/2), completed with 2 local objects.
To https://github.com/Anand-kumar-git/Project1.git
   46d7993..b1a4281  dev -> dev

ADMIN@DESKTOP-0E234HV MINGW64 ~/Project1 (dev)
○ $
```

15.Pipeline Build in dev Branch:

The screenshot displays the Jenkins web interface for a project named 'Project1' on the 'dev' branch. The interface includes a sidebar with navigation options: Status (selected), Changes, Build Now, View Configuration, Stages, and Pipeline Syntax. The main content area shows the 'dev' branch status with a green checkmark and the full project name 'Project1/dev'. Below this, there are 'Permalinks' for various build types: Last build (#4), Last stable build (#4), Last successful build (#4), Last failed build (#3), Last unsuccessful build (#3), and Last completed build (#4). On the left, a 'Builds' panel lists recent builds: #4 (9:15 PM, successful), #3 (9:01 PM, failed), #2 (8:57 PM, failed), and #1 (8:55 PM, failed).

16. Docker Image Pushed to Dockerhub dev Repository:

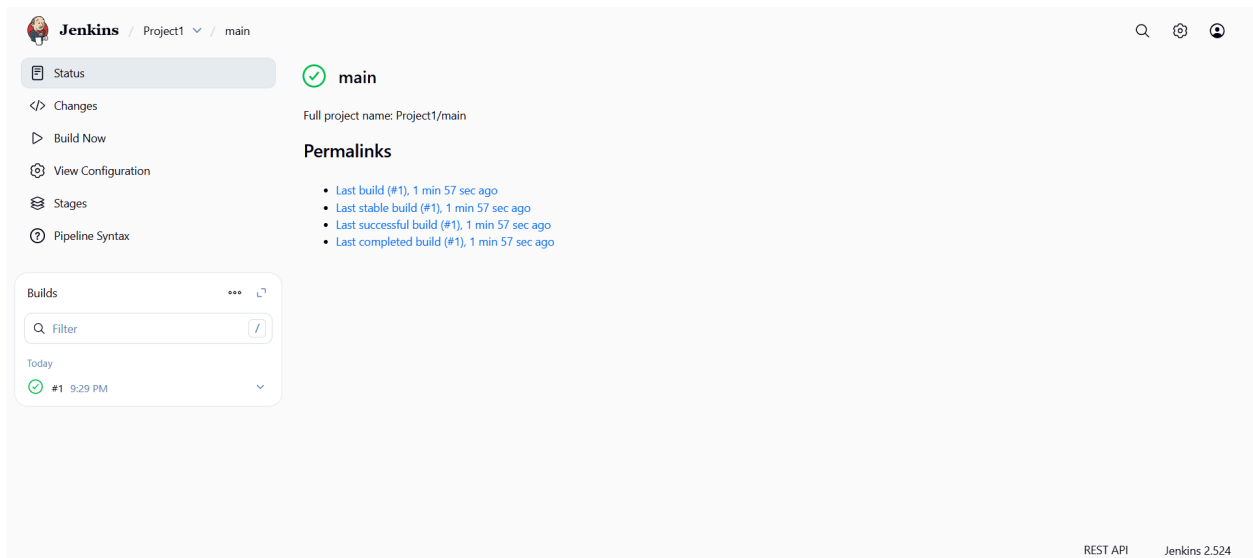
The screenshot shows the Docker Hub interface for a repository named 'anand20003/dev'. The repository is under the user 'anand20003' (Docker Personal). The 'General' tab is selected, showing the repository's details: 'Last pushed 9 minutes ago' and 'Repository size: 22.1 MB'. The 'Tags' section indicates 'This repository contains 0 tag(s)'. A table lists the tags, with one entry: 'latest' (OS: Image, Type: Image, Pulled: less than 1 day, Pushed: 9 minutes). The 'Image Management' tab is also visible, showing 'DOCKER SCOUT INACTIVE'. The 'Docker commands' section provides the command to push a new tag: 'docker push anand20003/dev:tagname'. The 'Build with Docker Build Cloud' section is also present, highlighting the benefits of using Docker Build Cloud for faster builds.

17. Merging dev to main:

```
ADMIN@DESKTOP-0E234HV MINGW64 ~/Project1 (main)
$ git merge dev
Updating f2fce62..b1a4281
Fast-forward
 .dockerignore | 9 ++++++
 .gitignore    | 1 +
 Dockerfile    | 9 ++++++
 Jenkinsfile   | 69 ++++++
 build.sh      | 5 +++++
 deploy.sh     | 10 ++++++
 6 files changed, 103 insertions(+)
 create mode 100644 .dockerignore
 create mode 100644 .gitignore
 create mode 100644 Dockerfile
 create mode 100644 Jenkinsfile
 create mode 100644 build.sh
 create mode 100644 deploy.sh

ADMIN@DESKTOP-0E234HV MINGW64 ~/Project1 (main)
$ git push origin main
```

18. Pipeline Build in Main branch:

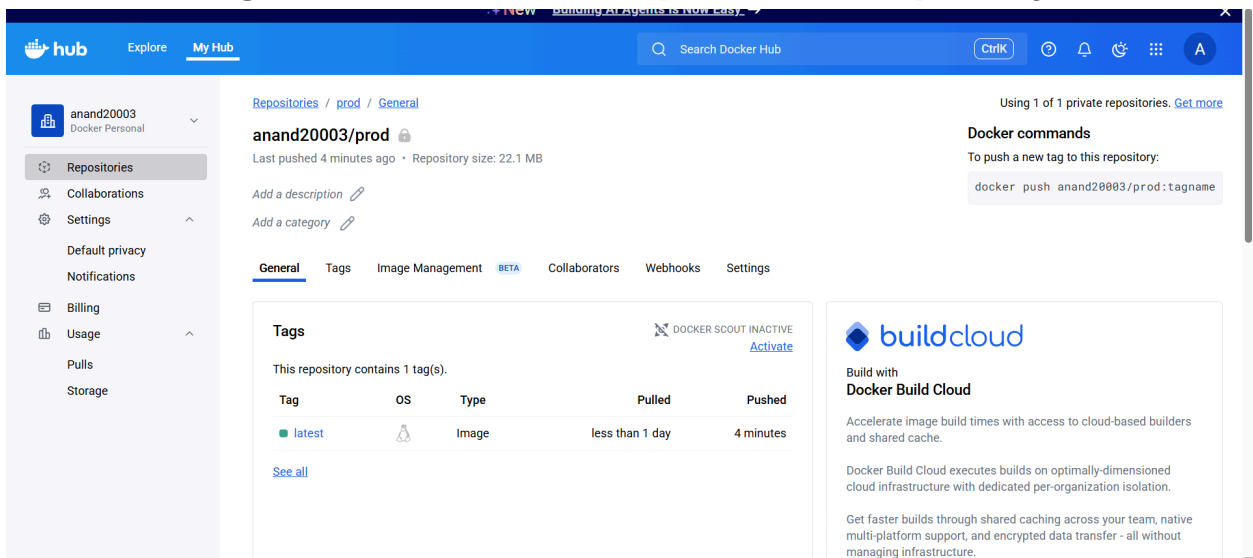


The screenshot shows the Jenkins web interface for a project named 'Project1' in the 'main' branch. The status is 'main' with a green checkmark. The left sidebar contains navigation links: Status, Changes, Build Now, View Configuration, Stages, and Pipeline Syntax. The main area displays 'Full project name: Project1/main' and a list of 'Permalinks' for the latest build. A 'Builds' table on the left shows a single build (#1) completed at 9:29 PM today.

Builds
Filter
Today
✓ #1 9:29 PM

REST API Jenkins 2.524

19. Docker Image Pushed to DockerHub prod Private Repository:



The screenshot shows the DockerHub interface for the 'anand20003/prod' repository. The left sidebar shows the user profile and repository settings. The main area displays the repository details, including the 'latest' tag pushed 4 minutes ago. A 'Docker commands' section shows the command to push a new tag. A 'buildcloud' advertisement is visible on the right.

Repositories / prod / General

anand20003/prod

Last pushed 4 minutes ago · Repository size: 22.1 MB

Add a description

Add a category

General Tags Image Management BETA Collaborators Webhooks Settings

Tags

This repository contains 1 tag(s).

Tag	OS	Type	Pulled	Pushed
latest	linux	Image	less than 1 day	4 minutes

See all

DOCKER SCOUT INACTIVE

Activate

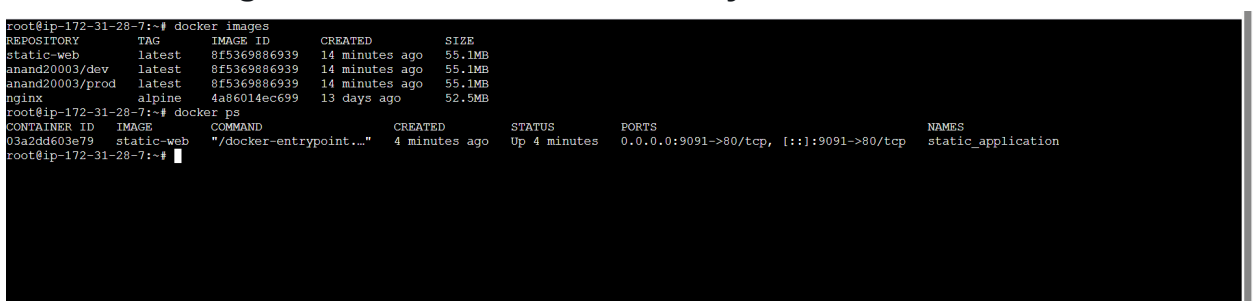
Build with Docker Build Cloud

Accelerate image build times with access to cloud-based builders and shared cache.

Docker Build Cloud executes builds on optimally-dimensioned cloud infrastructure with dedicated per-organization isolation.

Get faster builds through shared caching across your team, native multi-platform support, and encrypted data transfer - all without managing infrastructure.

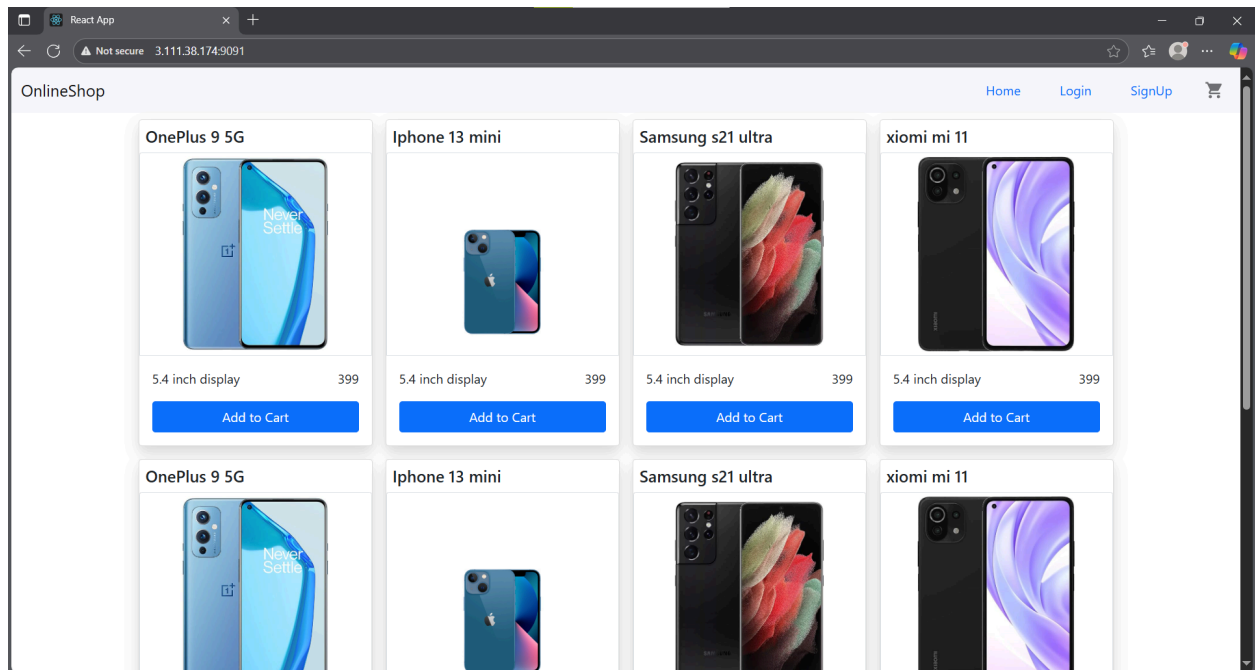
20. Docker Image and Container Build Locally:



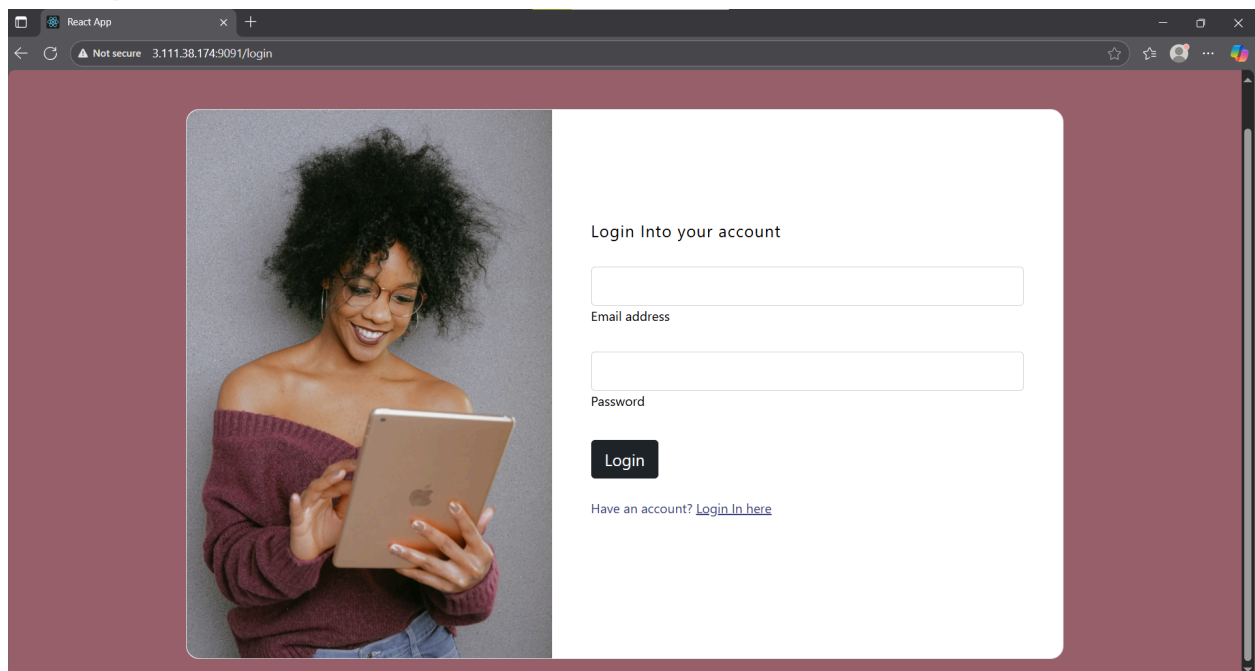
```
root@ip-172-31-28-7:~# docker images
REPOSITORY          TAG                 IMAGE ID            CREATED             SIZE
static-web          latest             8f5369886939       14 minutes ago     55.1MB
anand20003/dev      latest             8f5369886939       14 minutes ago     55.1MB
anand20003/prod     latest             8f5369886939       14 minutes ago     55.1MB
nginx               alpine             4a86014ec699       13 days ago        52.5MB

root@ip-172-31-28-7:~# docker ps
CONTAINER ID   IMAGE     COMMAND                  CREATED             STATUS              PORTS                               NAMES
03a2dd603e79   static-web "/docker-entrypoint..." 4 minutes ago       Up 4 minutes       0.0.0.0:9091->80/tcp, [::]:9091->80/tcp   static_application
```

21.Output 1:



22.Output 2:



23.Output 3:

React App

Not secure 3.111.38.174:9091/signup

OnlineShop

Home Login SignUp



Sign Up your account

User Name

Email address

Password

[Sign Up](#)

Have an account? [Login In here](#)

24.Output 4:

React App

Not secure 3.111.38.174:9091/cart

OnlineShop

Home Login SignUp

Shopping Cart

#	Product	Price	Quantity	Total
1	 OnePlus 9 5G 5.4 inch display Remove	\$399	- 1 +	\$399
1	 Iphone 13 mini 5.4 inch display Remove	\$399	- 1 +	\$399
1	 Samsung s21 ultra 5.4 inch display Remove	\$399	- 1 +	\$399

[Clear Cart](#)

SubTotal: \$1197

Taxes and Shipping Calculated

[Login to CheckOut](#)

[← Continue Shopping](#)

6.AWS:

1.Launching EC2 Instance with t2.micro:

The screenshot shows the 'Launch an instance' wizard in the AWS Management Console. The interface is divided into several sections:

- Name and tags:** A text input field contains 'Project1'. A link 'Add additional tags' is visible.
- Application and OS Images (Amazon Machine Image):** A search bar is present. Below it, a 'Quick Start' tab is active, showing a grid of AMIs including Amazon Linux, macOS, Ubuntu, Windows, Red Hat, SUSE Linux, and Debian. The 'Ubuntu' AMI is selected.
- Summary:** A sidebar on the right provides a summary of the configuration:
 - Number of instances:** 1
 - Software Image (AMI):** Amazon Linux 2023.8.2...read more
 - Virtual server type (instance type):** t2.micro
 - Firewall (security group):** New security group
 - Storage (volumes):** 1 volume(s) - 8 GiB
- Buttons:** At the bottom right, there are 'Cancel', 'Launch instance', and 'Preview code' buttons.

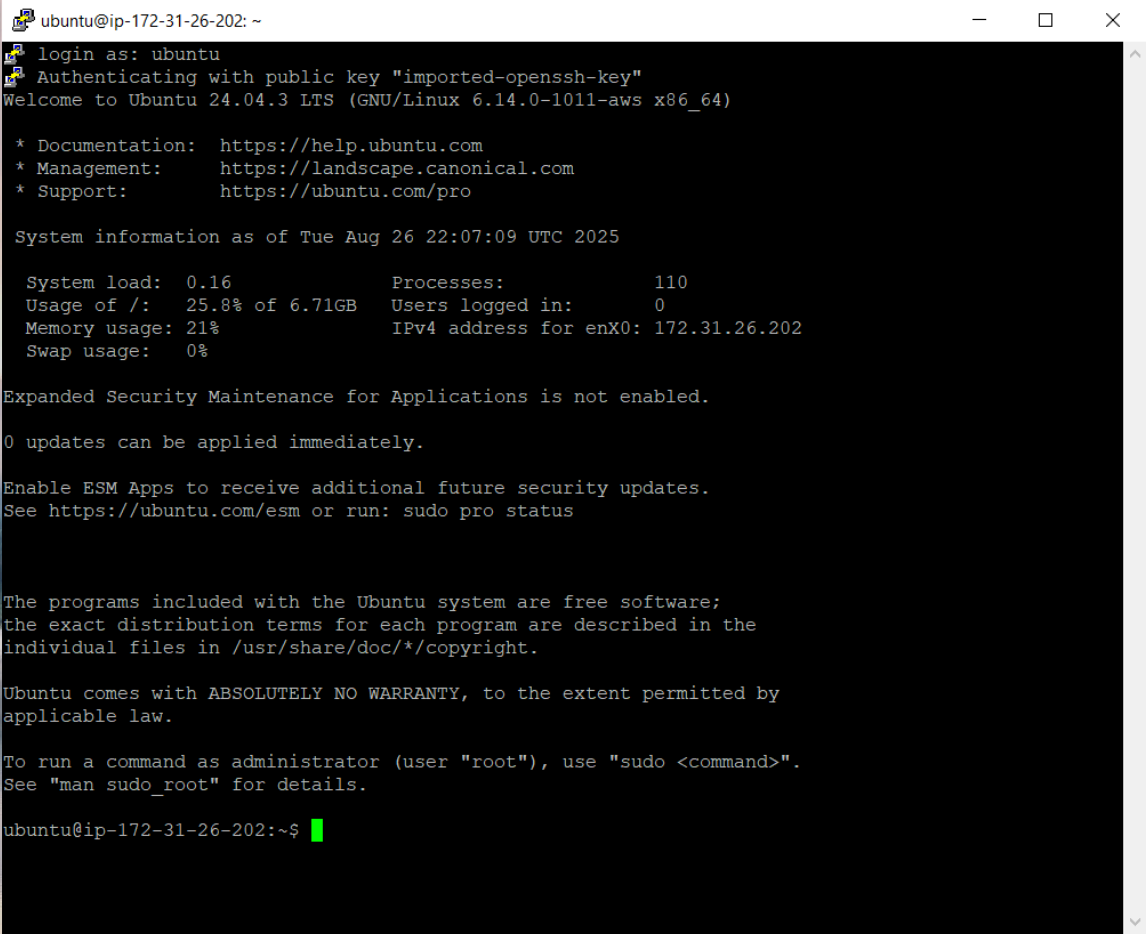
2. Security Group Configurations:

The screenshot shows the 'Edit inbound rules' page for a Security Group in the AWS Management Console. The page displays a table of existing inbound rules and an 'Add rule' button.

Security group rule ID	Type	Protocol	Port range	Source	Description - optional	Actions
sgr-0232081b632cb7f85	HTTPS	TCP	443	Custom		Delete
sgr-0d76c7fb6afdd8e1d	HTTP	TCP	80	Custom		Delete
sgr-00974c005bd504d8a	SSH	TCP	22	My IP	Login Access	Delete
-	Custom TCP	TCP	3000	Anyw...	Custom Application Port	Delete

Below the table, there is a warning message: "Rules with source of 0.0.0.0/0 or :::0 allow all IP addresses to access your instance. We recommend setting security group rules to allow access from known IP addresses only." At the bottom right, there are 'Cancel', 'Preview changes', and 'Save rules' buttons.

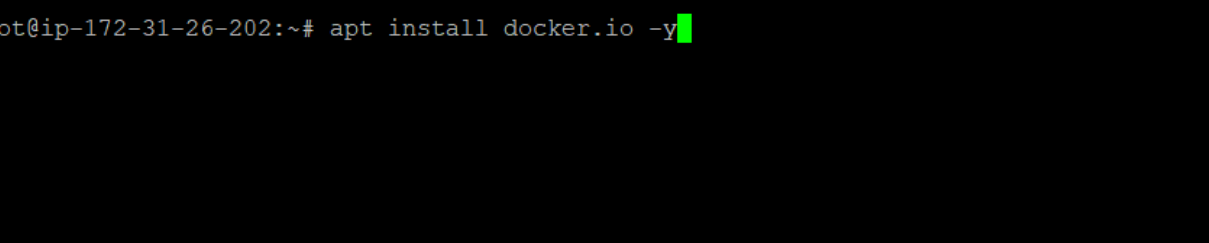
3.Login into the server with my IP:



A terminal window titled 'ubuntu@ip-172-31-26-202: ~' is displayed against a background image of a lake and mountains. The terminal shows the login process for the 'ubuntu' user, including authentication with a public key. It then displays system information as of Tuesday, August 26, 2025, at 22:07:09 UTC. The system load is 0.16, with 110 processes running. The root filesystem usage is 25.8% of 6.71GB, memory usage is 21%, and swap usage is 0%. It also shows that no updates can be applied immediately and provides links for documentation, management, and support. Finally, it displays the Ubuntu warranty disclaimer and instructions on how to run commands as administrator.

```
ubuntu@ip-172-31-26-202: ~  
login as: ubuntu  
Authenticating with public key "imported-openssh-key"  
Welcome to Ubuntu 24.04.3 LTS (GNU/Linux 6.14.0-1011-aws x86_64)  
  
* Documentation:  https://help.ubuntu.com  
* Management:    https://landscape.canonical.com  
* Support:       https://ubuntu.com/pro  
  
System information as of Tue Aug 26 22:07:09 UTC 2025  
  
System load:  0.16           Processes:            110  
Usage of /:   25.8% of 6.71GB Users logged in:          0  
Memory usage: 21%           IPv4 address for enX0: 172.31.26.202  
Swap usage:   0%  
  
Expanded Security Maintenance for Applications is not enabled.  
  
0 updates can be applied immediately.  
  
Enable ESM Apps to receive additional future security updates.  
See https://ubuntu.com/esm or run: sudo pro status  
  
The programs included with the Ubuntu system are free software;  
the exact distribution terms for each program are described in the  
individual files in /usr/share/doc/*/copyright.  
  
Ubuntu comes with ABSOLUTELY NO WARRANTY, to the extent permitted by  
applicable law.  
  
To run a command as administrator (user "root"), use "sudo <command>".  
See "man sudo_root" for details.  
  
ubuntu@ip-172-31-26-202:~$
```

4.Installing Docker:



A terminal window titled 'root@ip-172-31-26-202: ~' is shown. The user is logged in as 'root'. The command 'apt install docker.io -y' has been entered and is highlighted with a green cursor.

```
root@ip-172-31-26-202: ~  
root@ip-172-31-26-202:~# apt install docker.io -y
```

5.Docker Installed:

 root@ip-172-31-26-202: ~

```
root@ip-172-31-26-202:~# docker --version
Docker version 27.5.1, build 27.5.1-0ubuntu3~24.04.2
root@ip-172-31-26-202:~#
```

6.DockerHub Login Done:

```
root@ip-172-31-26-202:~# docker login

USING WEB-BASED LOGIN
To sign in with credentials on the command line, use 'docker login -u <username>'

Your one-time device confirmation code is: CGLG-JJKQ
Press ENTER to open your browser or submit your device code here: https://login.docker.com/activate

Waiting for authentication in the browser...
WARNING! Your password will be stored unencrypted in /root/.docker/config.json.
Configure a credential helper to remove this warning. See
https://docs.docker.com/engine/reference/commandline/login/#credential-stores

Login Succeeded
root@ip-172-31-26-202:~#
```

7. Pulling Docker Image From Dockerhub:

 root@ip-172-31-26-202: ~

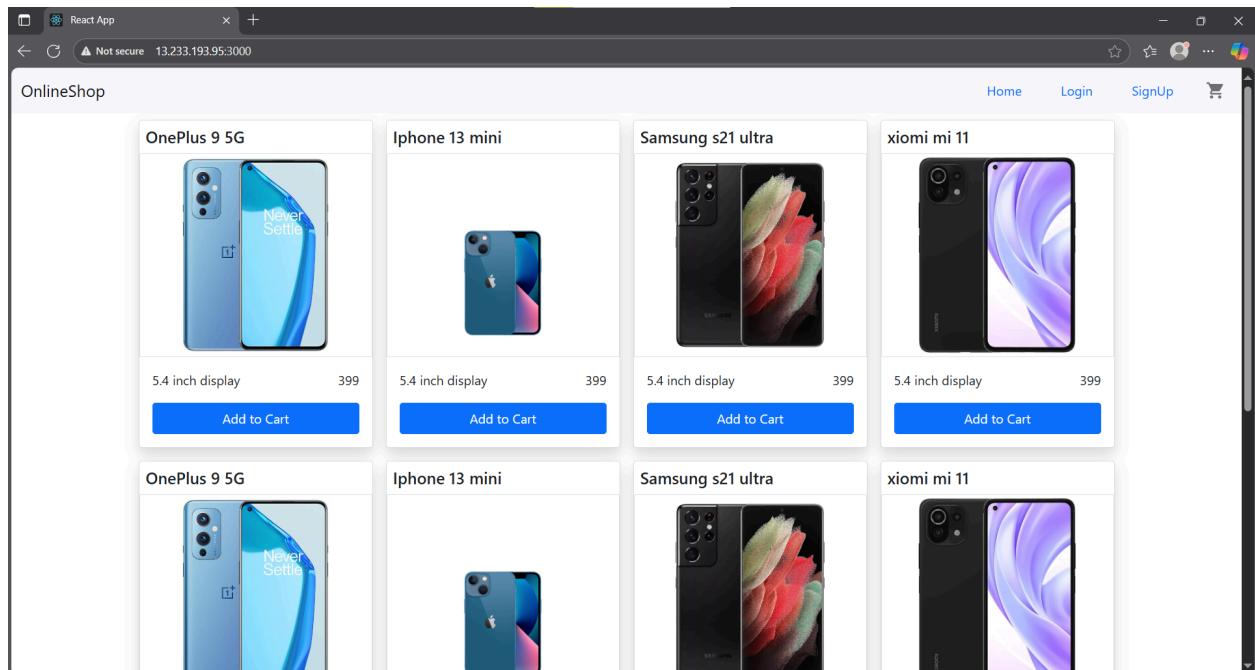
```
root@ip-172-31-26-202:~# docker pull anand20003/prod:latest
latest: Pulling from anand20003/prod
9824c27679d3: Pull complete
6bc572a340ec: Pull complete
403e3f251637: Pull complete
9adfb9ae99cb7: Pull complete
7a8a46741e18: Pull complete
c9ebe2ff2d2c: Pull complete
a992fbc61ecc: Pull complete
cb1ff4086f82: Pull complete
7252e8e57291: Pull complete
8e6b521f19db: Pull complete
Digest: sha256:f11b94cb4989c2d6062c45385ab49c49163fdadfe2752a3ad7ac1df6995f7c64
Status: Downloaded newer image for anand20003/prod:latest
docker.io/anand20003/prod:latest
root@ip-172-31-26-202:~#
```

8. Container Created With Pulled Image:

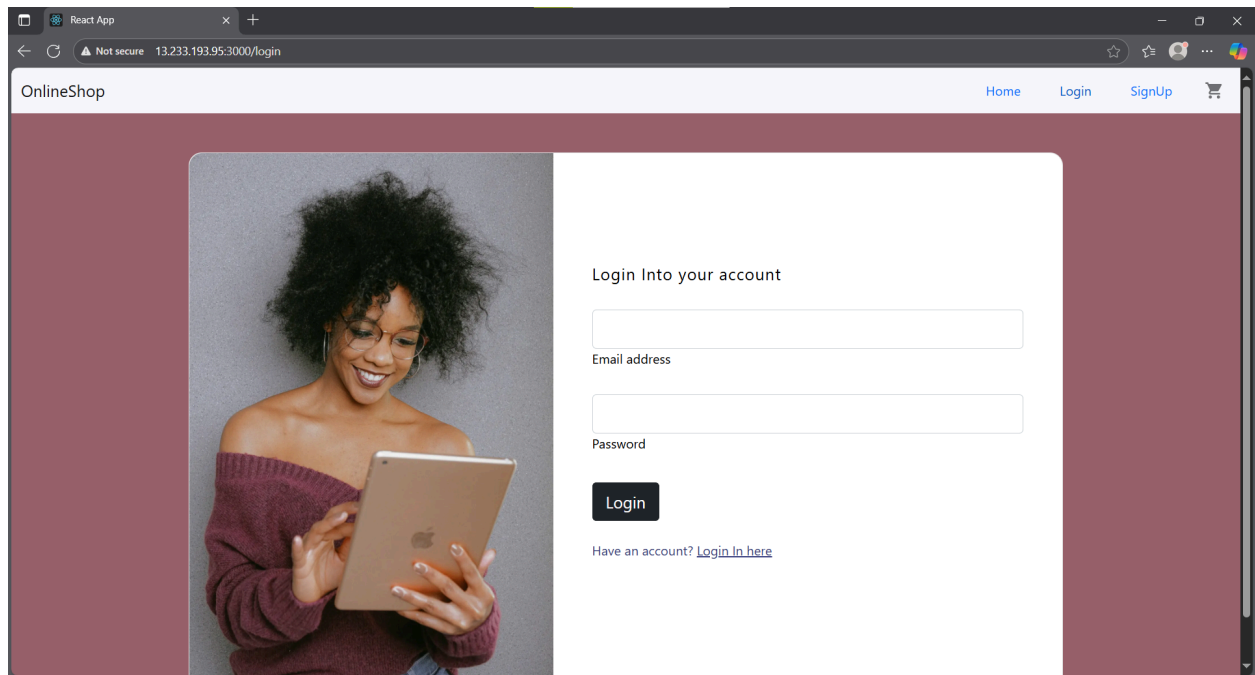
 root@ip-172-31-26-202: ~

```
root@ip-172-31-26-202:~# docker images
REPOSITORY          TAG          IMAGE ID          CREATED           SIZE
anand20003/prod     latest      8f5369886939     About an hour ago 55.1MB
root@ip-172-31-26-202:~# docker run -itd -p "3000:80" --name Reactjs_Application anand20003/prod
44455847f161ad69616edcb1ff9dd1cb03783418d542415e56314fc89d027d32
root@ip-172-31-26-202:~#
```

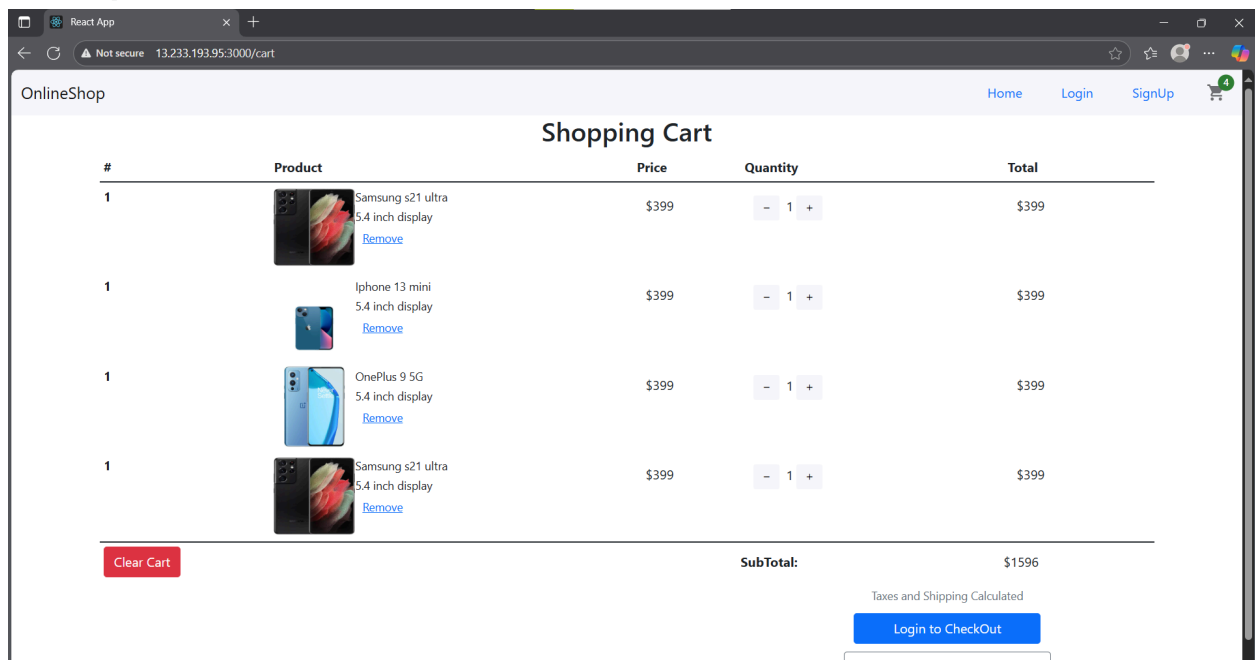

9.Output 1:



10.Output 2:

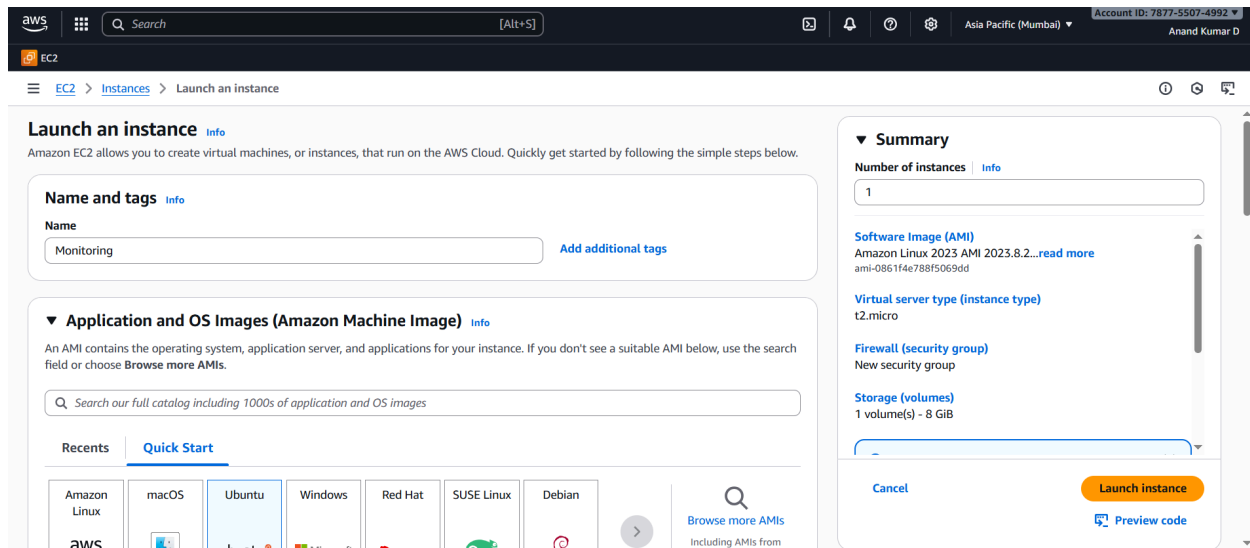


11.Output 3:



7. Monitoring:

1. Launching Instance For Monitoring:



2. Installing Docker-compose:

```
root@ip-172-31-23-1:~# apt install docker-compose
```

3. Docker-compose.yml file for Uptime Kuma:

```
GNU nano 7.2 docker-compose.yml
version: '3'
services:
  uptime-kuma:
    image: louislam/uptime-kuma:1
    container_name: uptime-kuma
    restart: always
    ports:
      - "3001:3001"
    volumes:
      - ./data:/app/data
```

4. Running Docker-compose File:

```
root@ip-172-31-23-1:~# docker-compose up -d
Creating network "root_default" with the default driver
Pulling uptime-kuma (louislam/uptime-kuma:1)...
1: Pulling from louislam/uptime-kuma
b338562f40a7: Pull complete
874bf4d93720: Pull complete
b16337721583: Pull complete
7d955db85b85: Pull complete
2c706596bd17: Pull complete
88a5c59ed14f: Pull complete
5a1d0a896c33: Pull complete
e68c2f25b946: Pull complete
2e6c90f010d6: Pull complete
ff15b10fabb8: Pull complete
4f4fb700ef54: Pull complete
d2a400cc8adb: Pull complete
Digest: sha256:431fee3be822b04861cf0e35daf4beef6b7cb37391c5f26c3ad6e12ce280fe18
Status: Downloaded newer image for louislam/uptime-kuma:1
Creating uptime-kuma ... done
root@ip-172-31-23-1:~#
```

5. Enabling Port 3001 to access Uptime-Kuma for Monitoring:

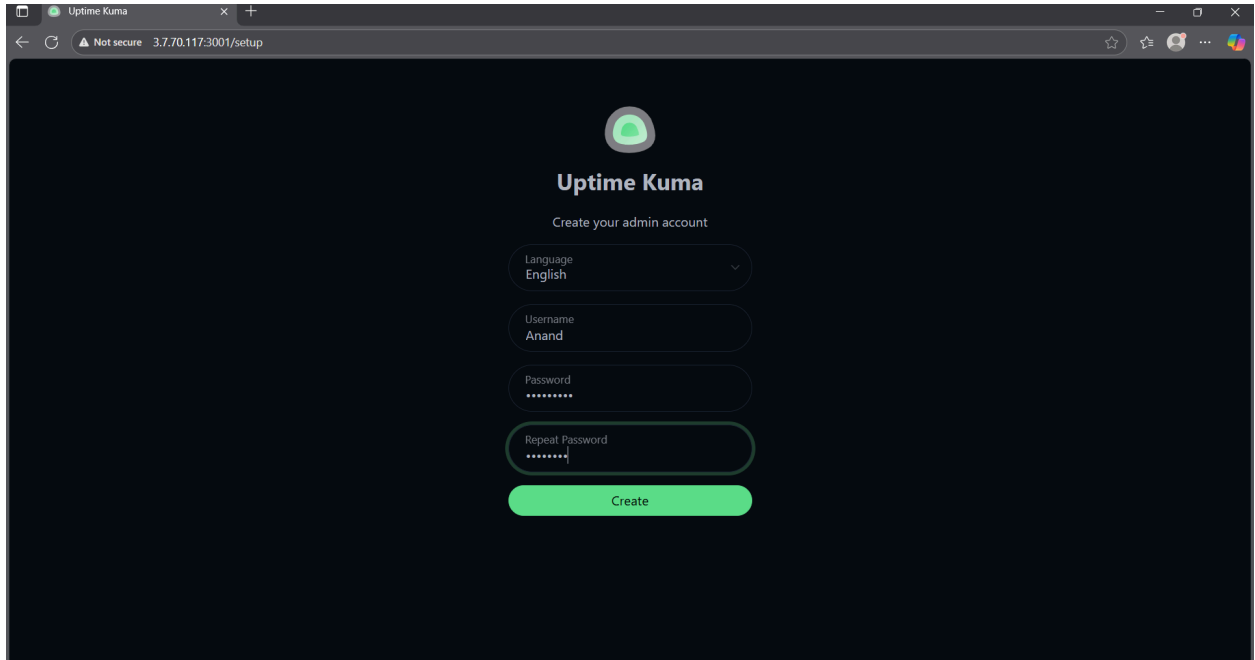
EC2 > Security Groups > sg-081a02413cad936a - launch-wizard-11 > Edit inbound rules

Edit inbound rules Info

Inbound rules control the incoming traffic that's allowed to reach the instance.

Security group rule ID	Type Info	Protocol Info	Port range Info	Source Info	Description - optional Info		
sg-0fc6d94bce5a0203a	HTTP	TCP	80	Custom	Q	0.0.0.0 X	Delete
sg-058e1012a600291df	SSH	TCP	22	Custom	Q	0.0.0.0 X	Delete
sg-0830b1dc228de0203	HTTPS	TCP	443	Custom	Q	0.0.0.0 X	Delete
sg-0fb0af36272c87dfb	Custom TCP	TCP	3001	Custom	Q	0.0.0.0 X	Delete
						0.0.0.0 X	

6. Creating Account in Uptime-Kuma for Monitoring:



The screenshot shows the Uptime Kuma setup page in a web browser. The page has a dark theme and a green logo at the top. The title "Uptime Kuma" is centered, followed by the subtitle "Create your admin account". Below this, there are four input fields: "Language" (set to "English"), "Username" (set to "Anand"), "Password" (masked with dots), and "Repeat Password" (masked with dots). A green "Create" button is at the bottom.

Uptime Kuma

Create your admin account

Language
English

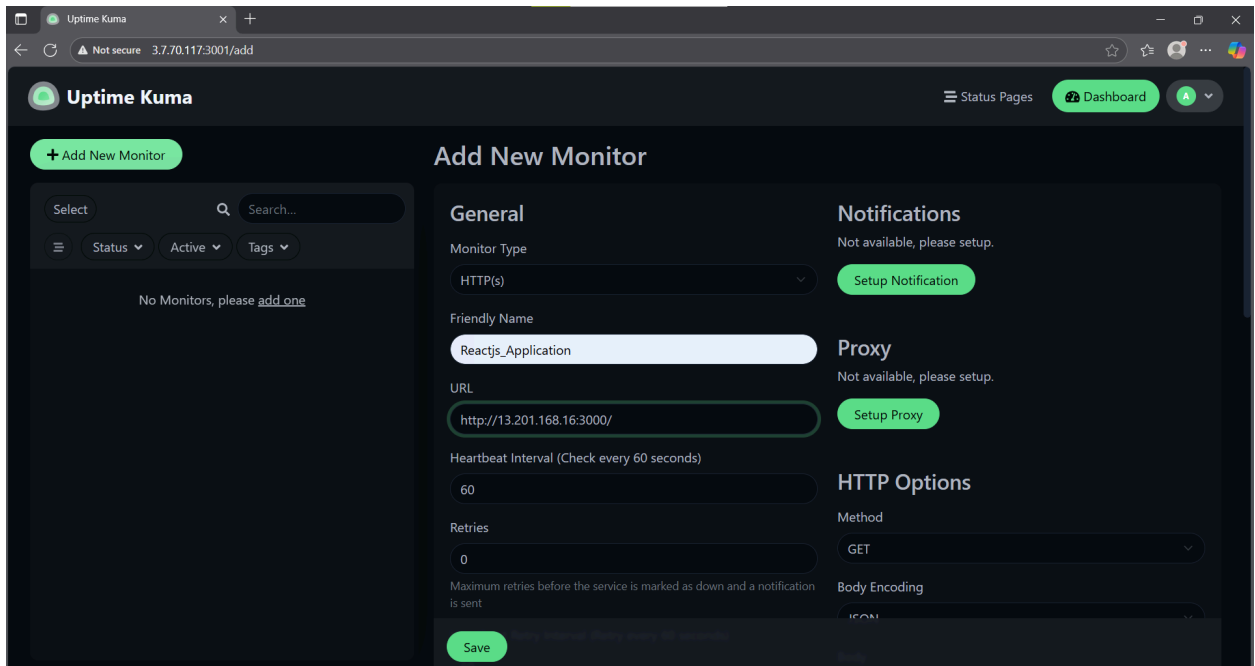
Username
Anand

Password

Repeat Password

Create

7. Adding New Monitor:



The screenshot shows the "Add New Monitor" page in the Uptime Kuma dashboard. The page has a dark theme and a green logo at the top. The title "Add New Monitor" is centered. On the left, there is a sidebar with a search bar and a list of monitors. The main content area is divided into three sections: "General", "Notifications", and "HTTP Options".

Uptime Kuma

Status Pages Dashboard

+ Add New Monitor

Select Search...

Status Active Tags

No Monitors, please [add one](#)

Add New Monitor

General

Monitor Type
HTTP(s)

Friendly Name
Reactjs_Application

URL
http://13.201.168.16:3000/

Heartbeat Interval (Check every 60 seconds)
60

Retries
0

Maximum retries before the service is marked as down and a notification is sent

Save

Notifications

Not available, please setup.

Setup Notification

Proxy

Not available, please setup.

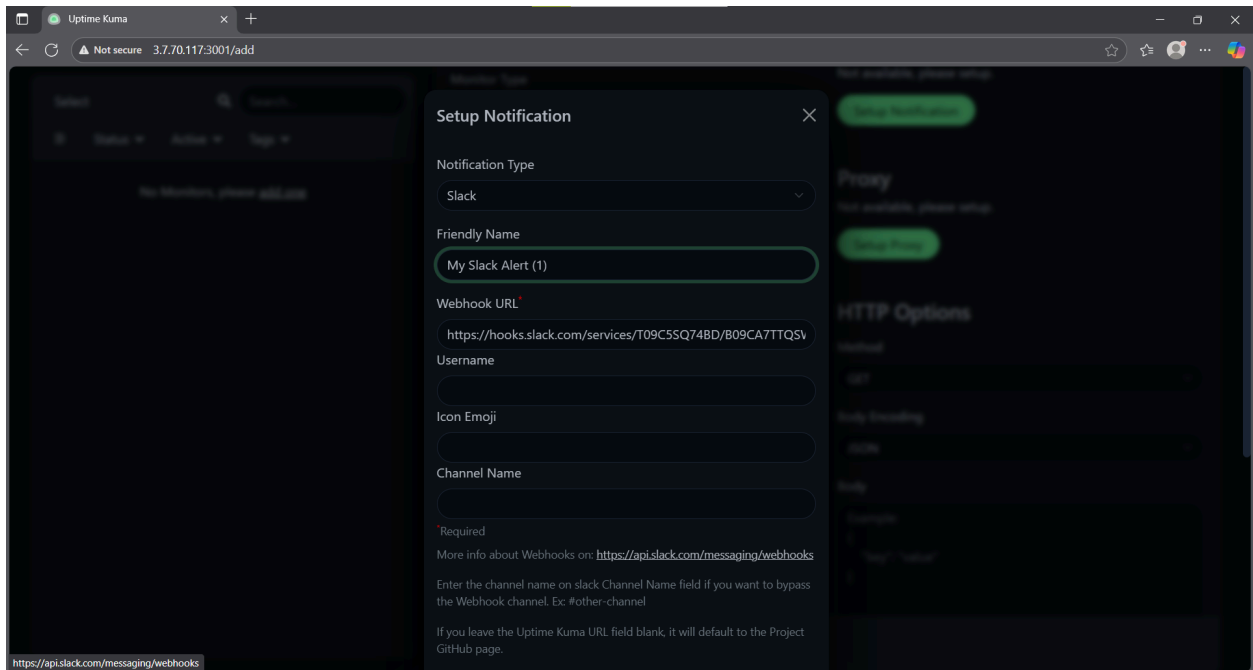
Setup Proxy

HTTP Options

Method
GET

Body Encoding
JSON

8. Setting up Slack Notification:



The screenshot shows the 'Setup Notification' modal in the Uptime Kuma web interface. The modal is titled 'Setup Notification' and has a close button (X) in the top right corner. It contains the following fields and options:

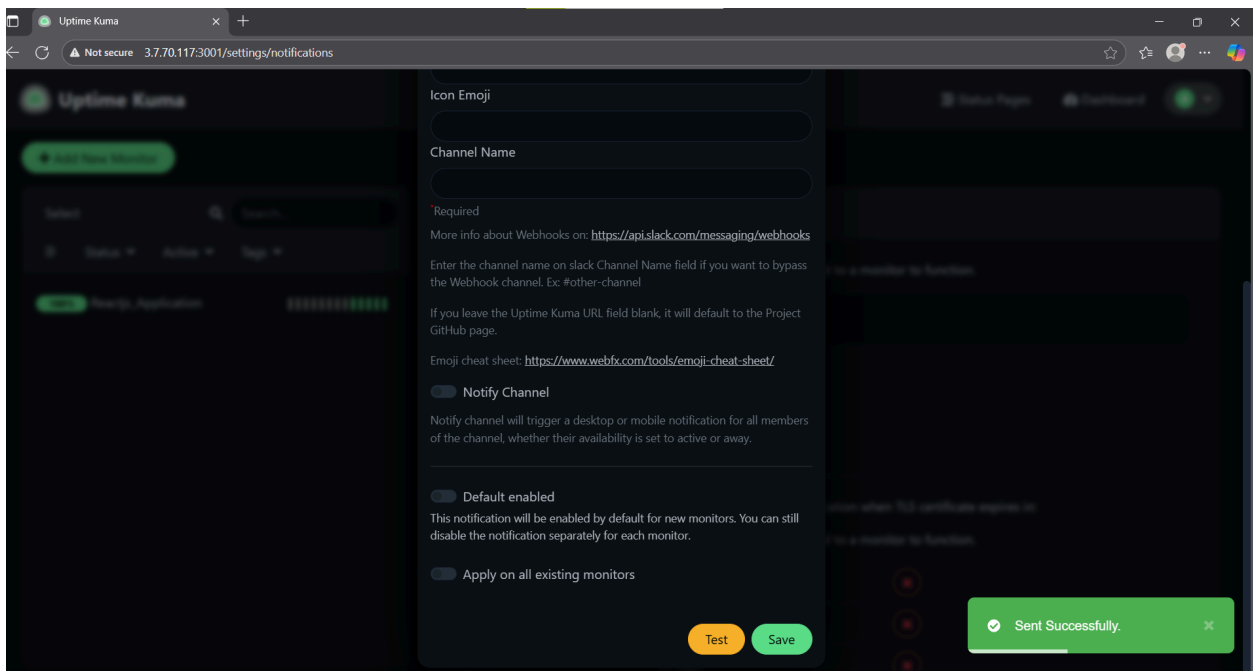
- Notification Type:** A dropdown menu with 'Slack' selected.
- Friendly Name:** A text input field containing 'My Slack Alert (1)'.
- Webhook URL:** A text input field containing 'https://hooks.slack.com/services/T09C5SQ74BD/B09CA7TTQSV'.
- Username:** An empty text input field.
- Icon Emoji:** An empty text input field.
- Channel Name:** An empty text input field.

Below the input fields, there is a 'Required' section with the following text:

More info about Webhooks on: <https://api.slack.com/messaging/webhooks>
Enter the channel name on slack Channel Name field if you want to bypass the Webhook channel. Ex: #other-channel
If you leave the Uptime Kuma URL field blank, it will default to the Project GitHub page.

The background of the modal shows a blurred view of the Uptime Kuma dashboard, including a 'Status' page and a 'Proxy' page.

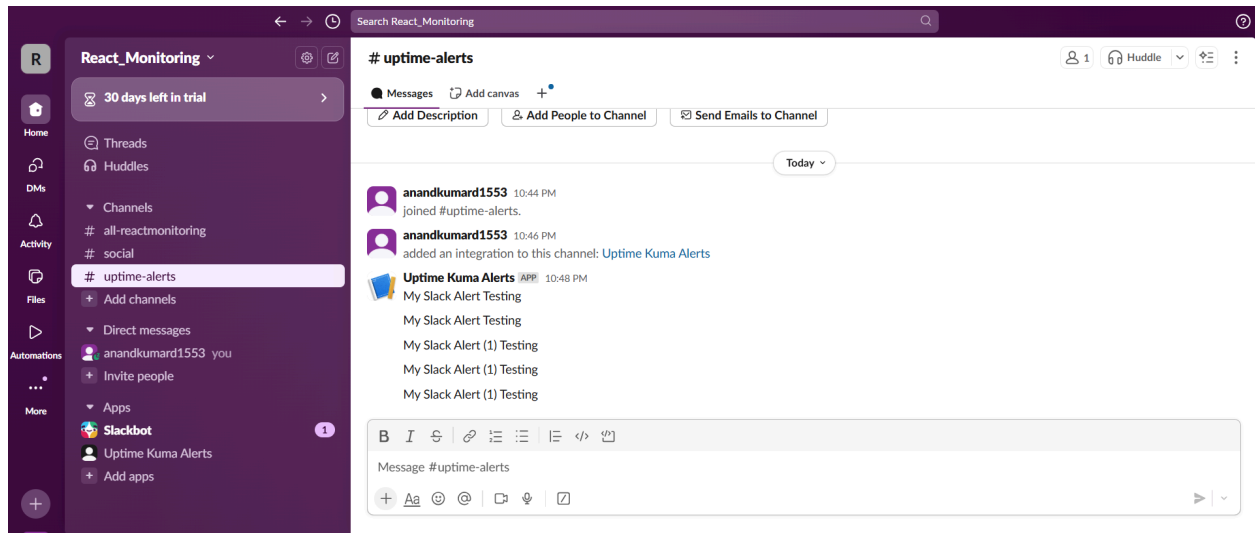
9. Testing the Notification:



The screenshot shows the 'Test' button in the Uptime Kuma web interface. The button is orange and labeled 'Test'. It is located at the bottom of the 'Setup Notification' modal. To the right of the button, there is a green message box that says 'Sent Successfully.' with a close button (X) in the top right corner.

The background of the modal shows a blurred view of the Uptime Kuma dashboard, including a 'Status' page and a 'Proxy' page.

10.Notification Received in Slack:



11.Monitor Dashboard in Uptime Kuma:

